



US 20250287573A1

(19) **United States**

(12) **Patent Application Publication**
Kim

(10) **Pub. No.: US 2025/0287573 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **SEMICONDUCTOR DEVICE**

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(21) Appl. No.: **19/008,323**

(22) Filed: **Jan. 2, 2025**

(30) **Foreign Application Priority Data**

Mar. 8, 2024 (KR) 10-2024-0033144

Publication Classification

(51) **Int. Cl.**
H10B 12/00 (2023.01)

(52) **U.S. Cl.**

CPC **H10B 12/485** (2023.02); **H10B 12/02**
(2023.02)

(57)

ABSTRACT

A semiconductor device may include a first insulating interlayer on a substrate, first contact plugs passing through the first insulating interlayer, a second insulating interlayer on the first insulating interlayer and first contact plugs, and first conductive patterns passing through the second insulating interlayer and extending below an upper surface of the first insulating interlayer in a vertical direction. Each first contact plug may include a first metal pattern and a first barrier metal pattern covering a portion of a sidewall of the first metal pattern, further covering a bottom of the first metal pattern, and further exposing an upper sidewall of the first metal pattern. Each of the first conductive patterns may have a first bottom surface contacting a portion of an upper surface of a respective first contact plug, and a second bottom surface contacting the first insulating interlayer and being lower than the first bottom surface.

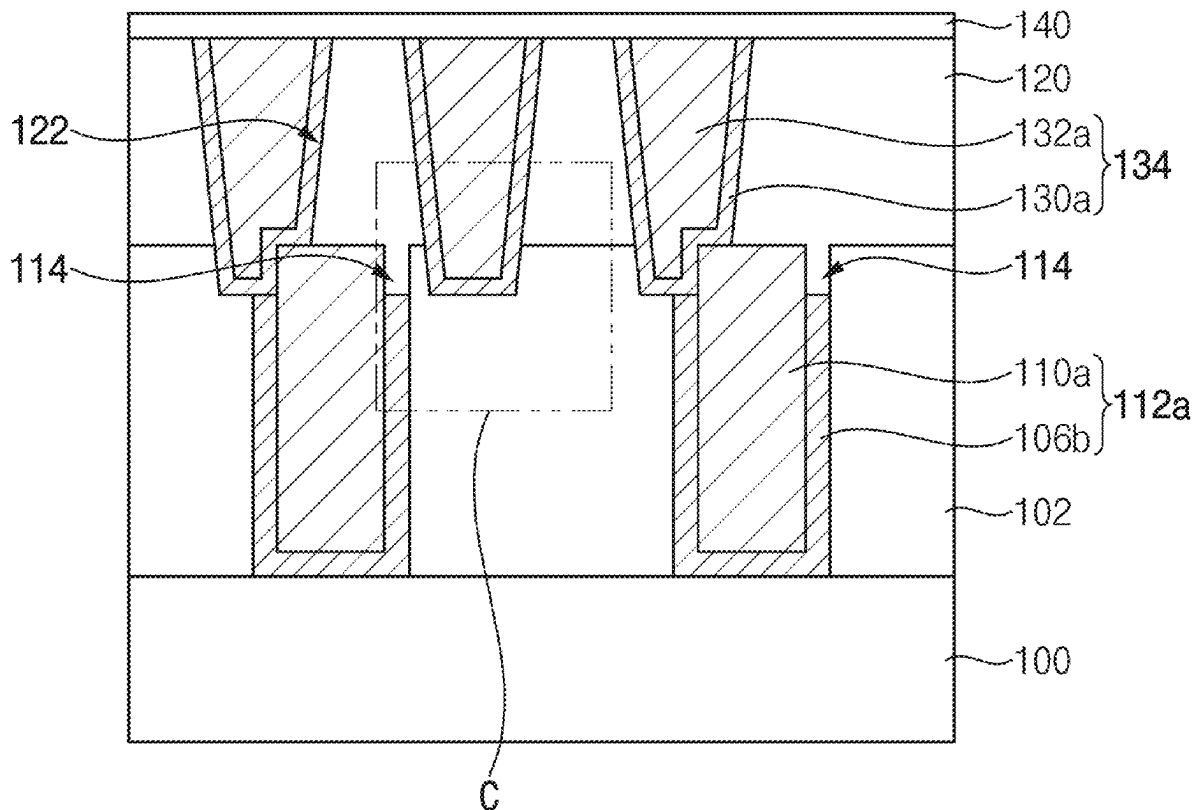


FIG. 1A

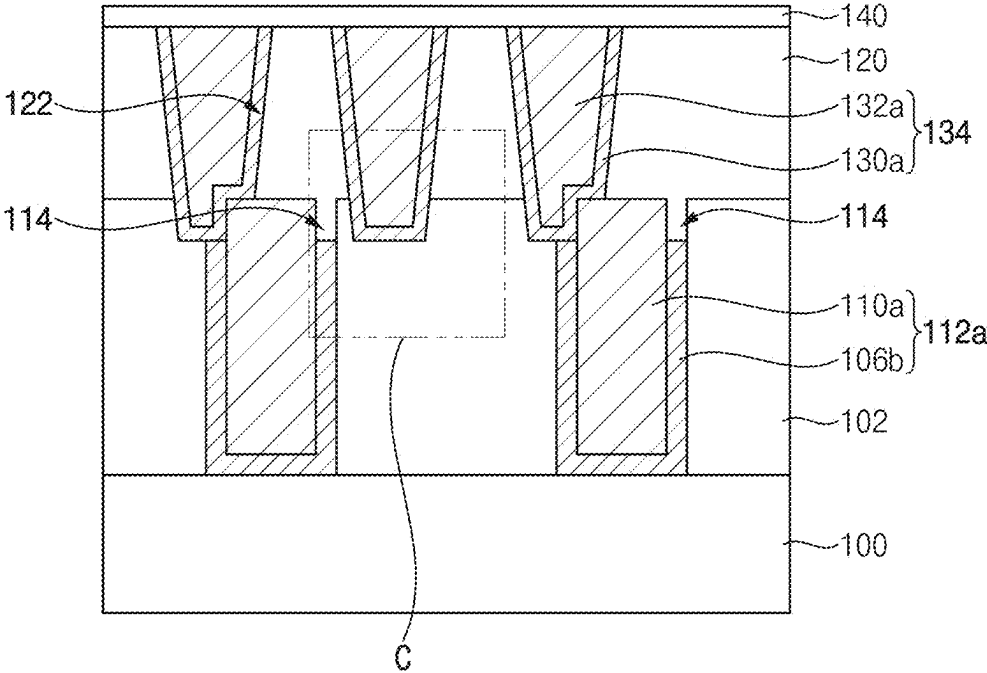


FIG. 1B

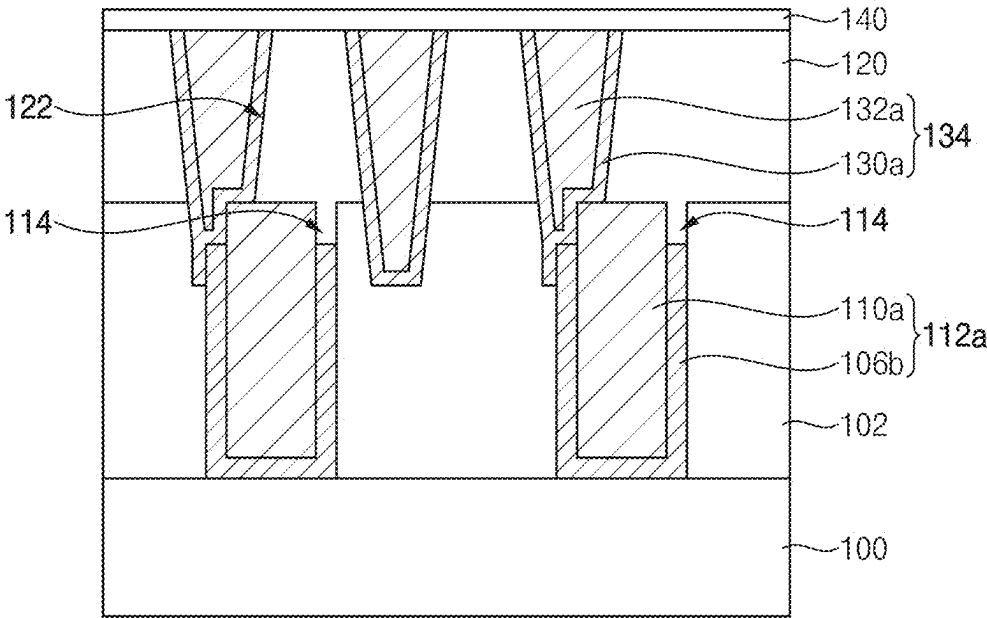


FIG. 1C

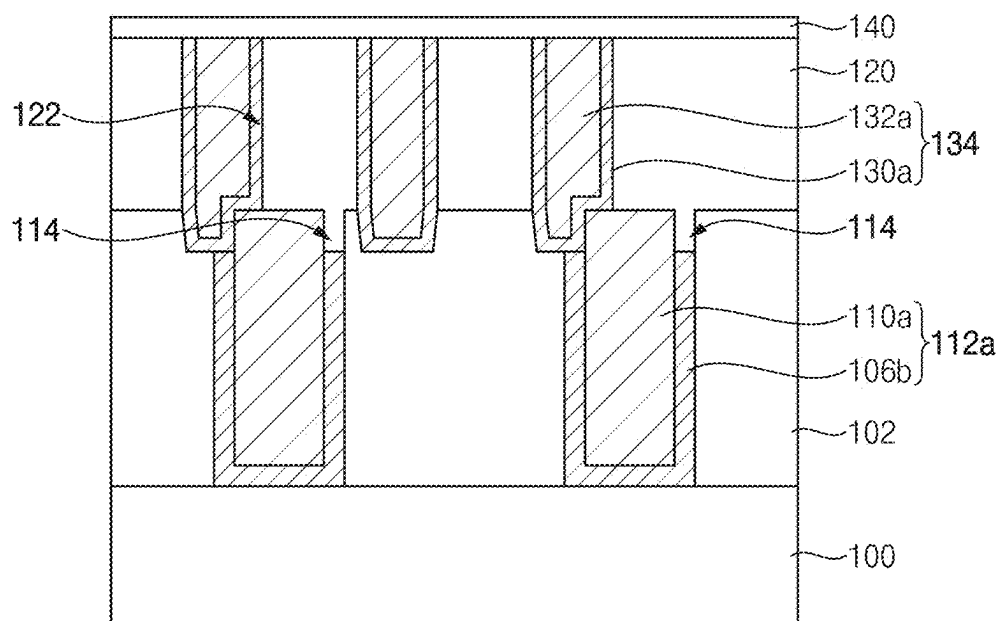


FIG. 2

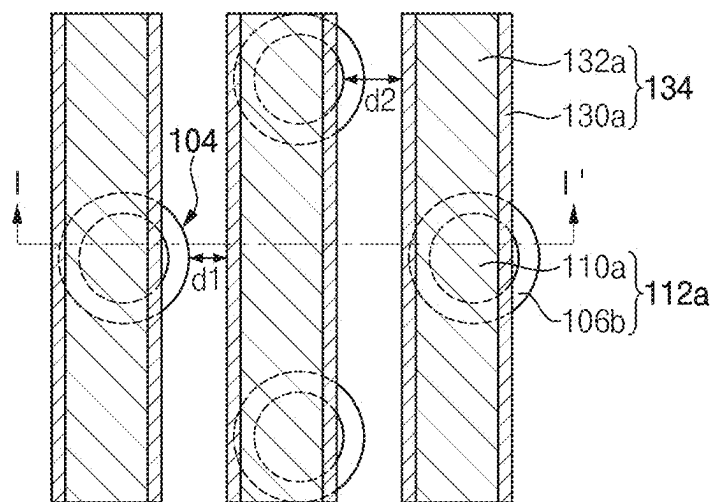


FIG. 3

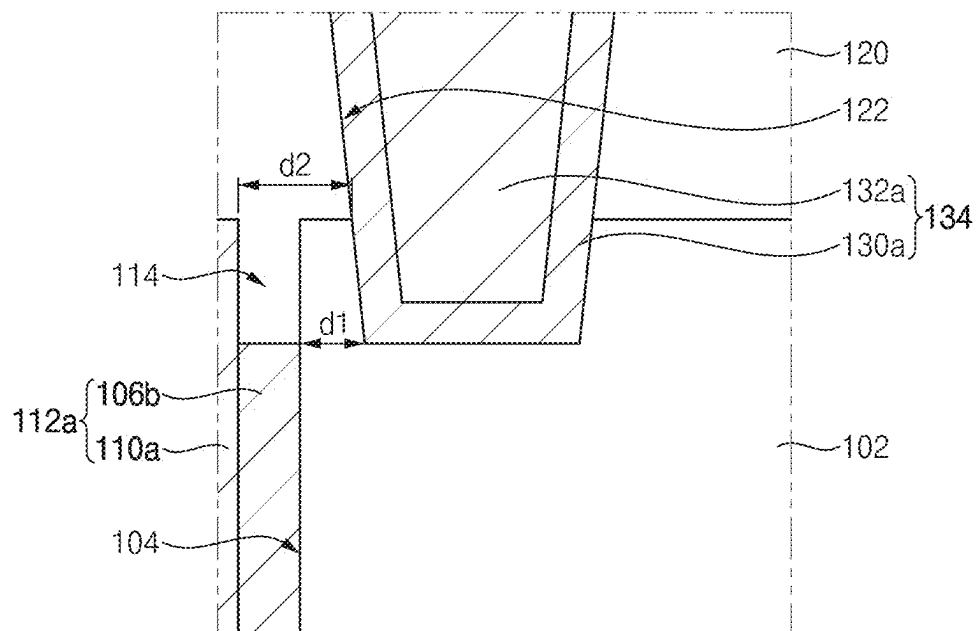


FIG. 4

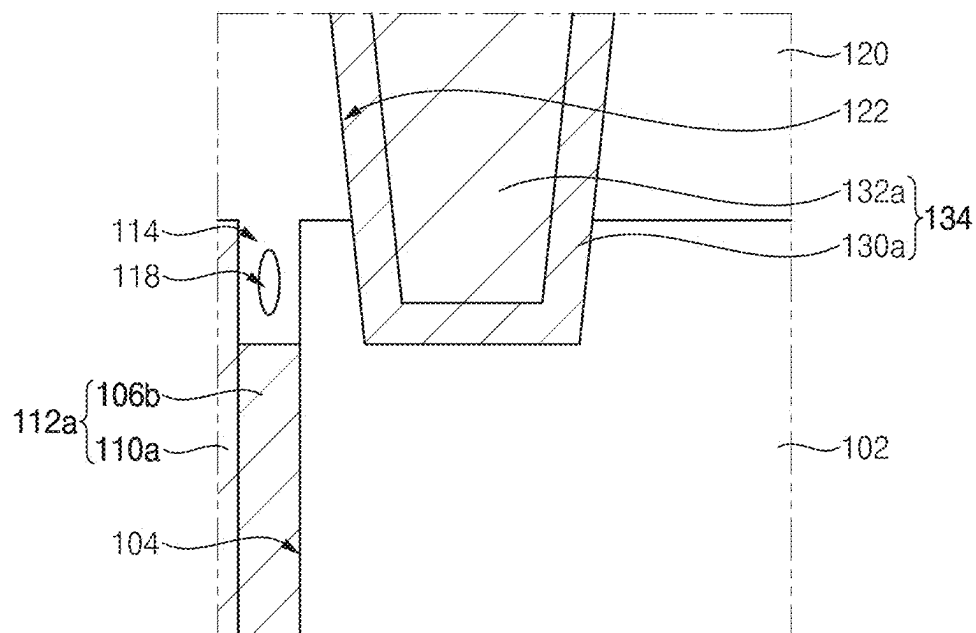


FIG. 5

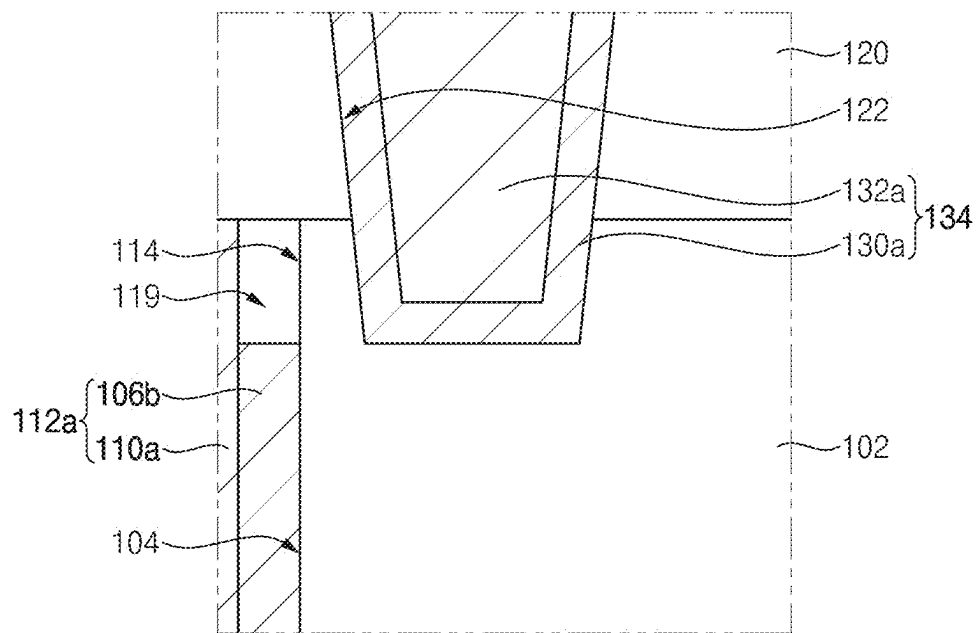


FIG. 6

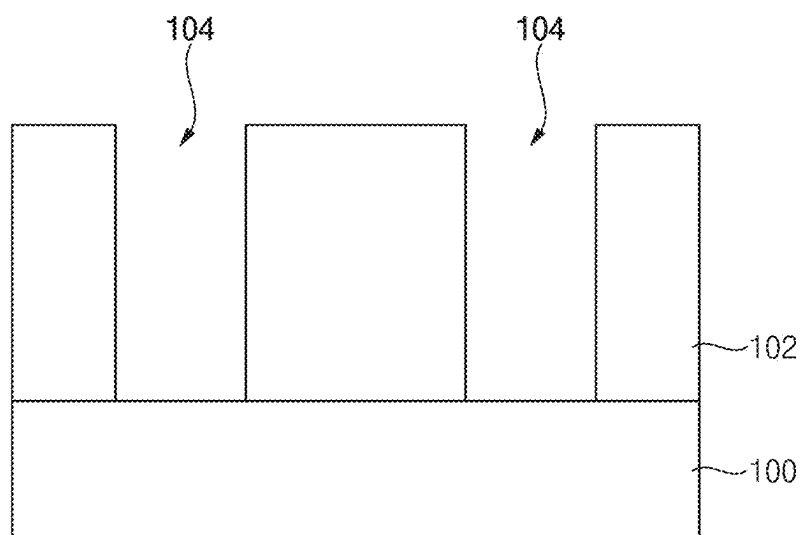


FIG. 7

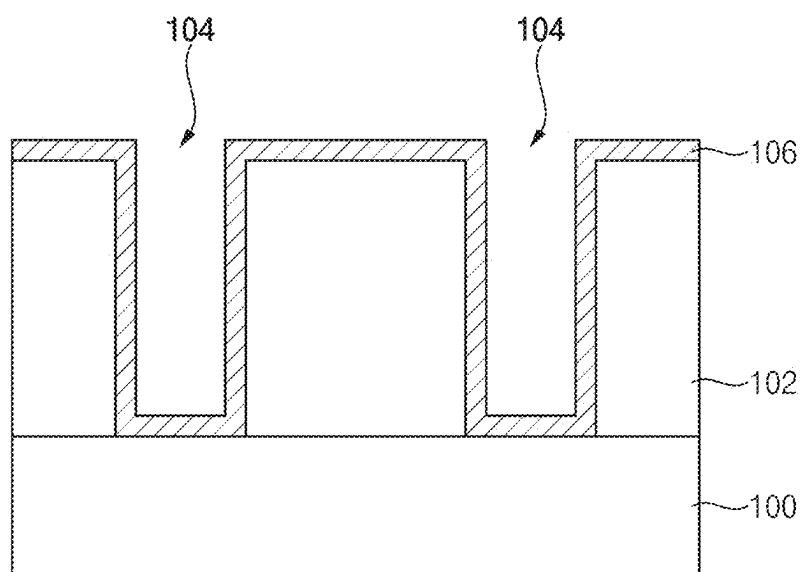


FIG. 8

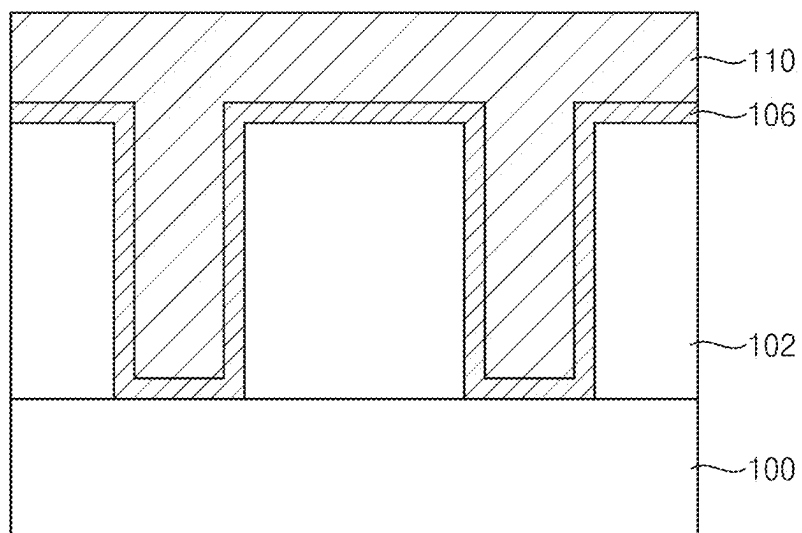


FIG. 9

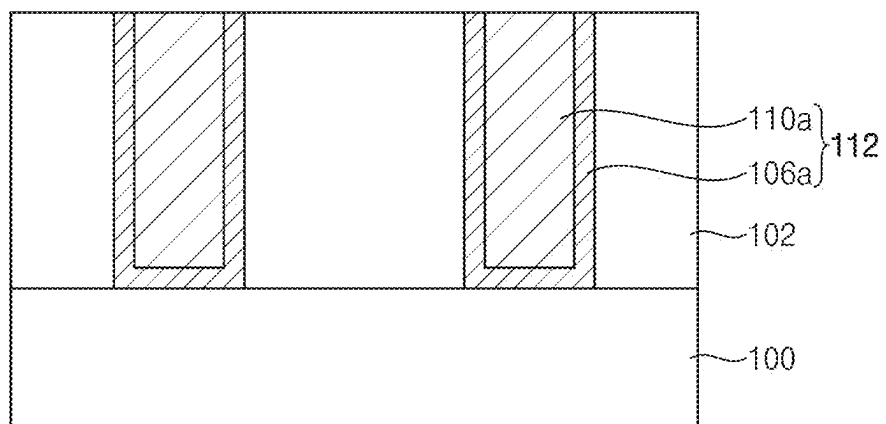


FIG. 10

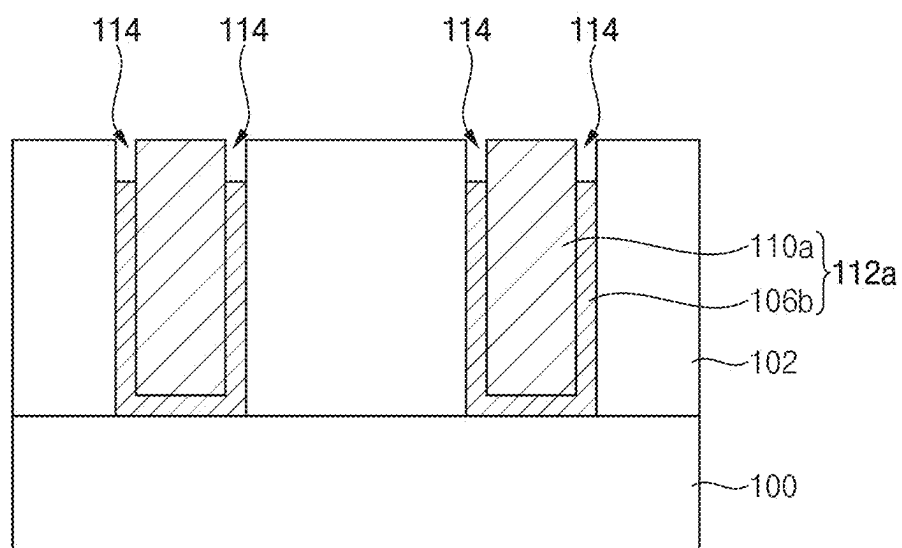


FIG. 11

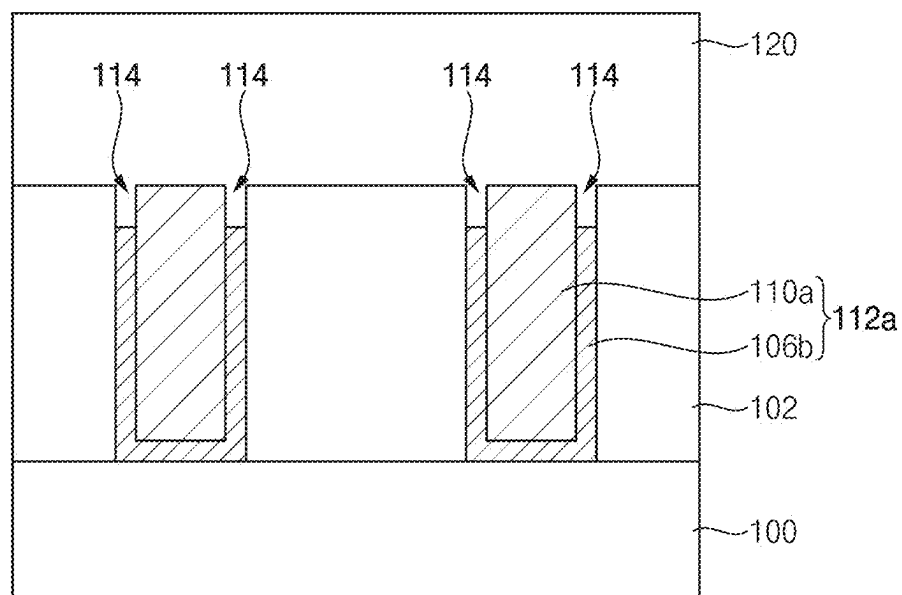


FIG. 12

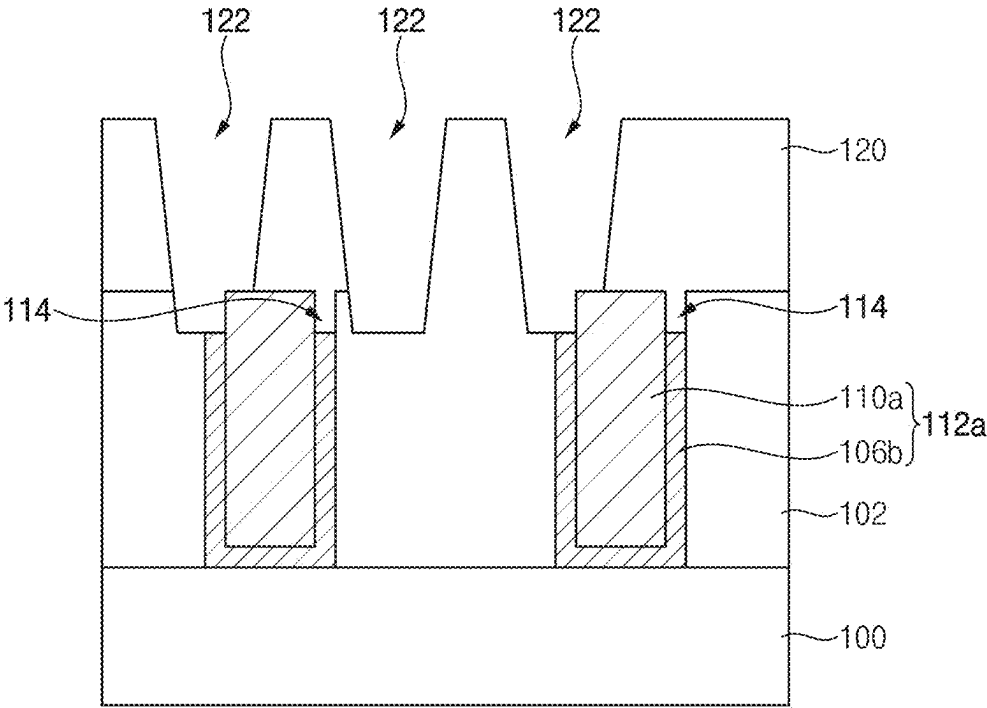


FIG. 13

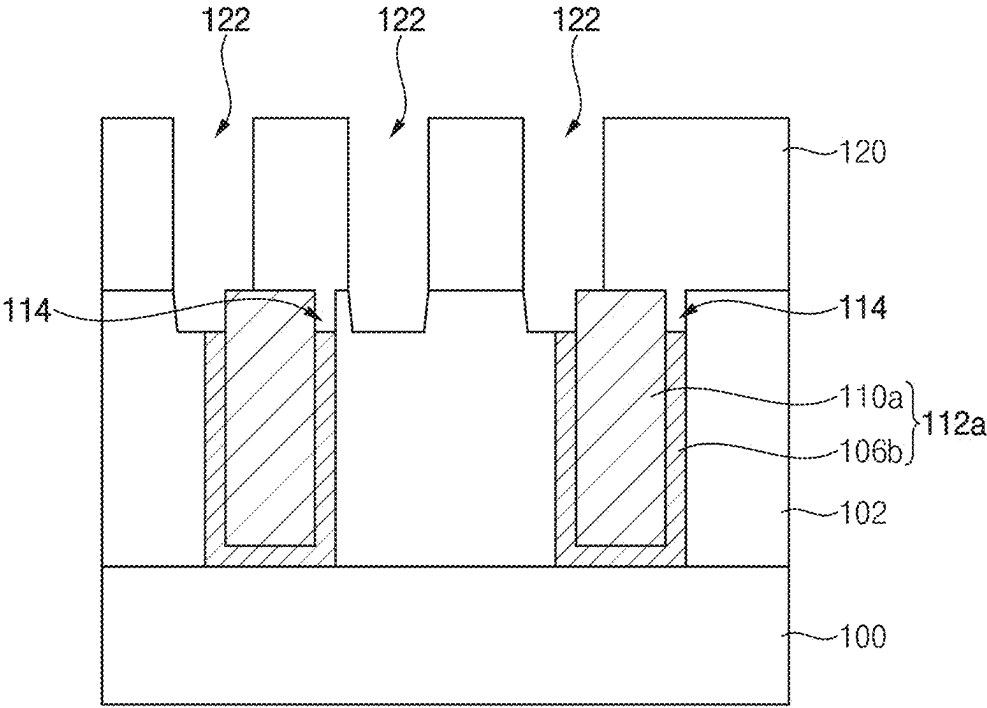


FIG. 14

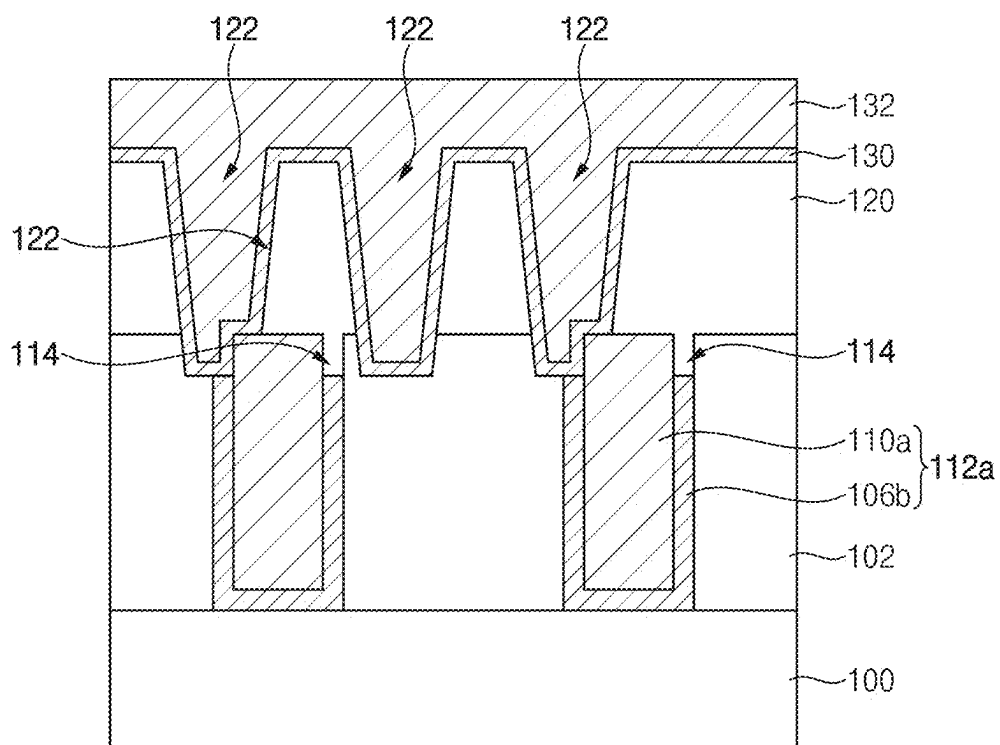


FIG. 15

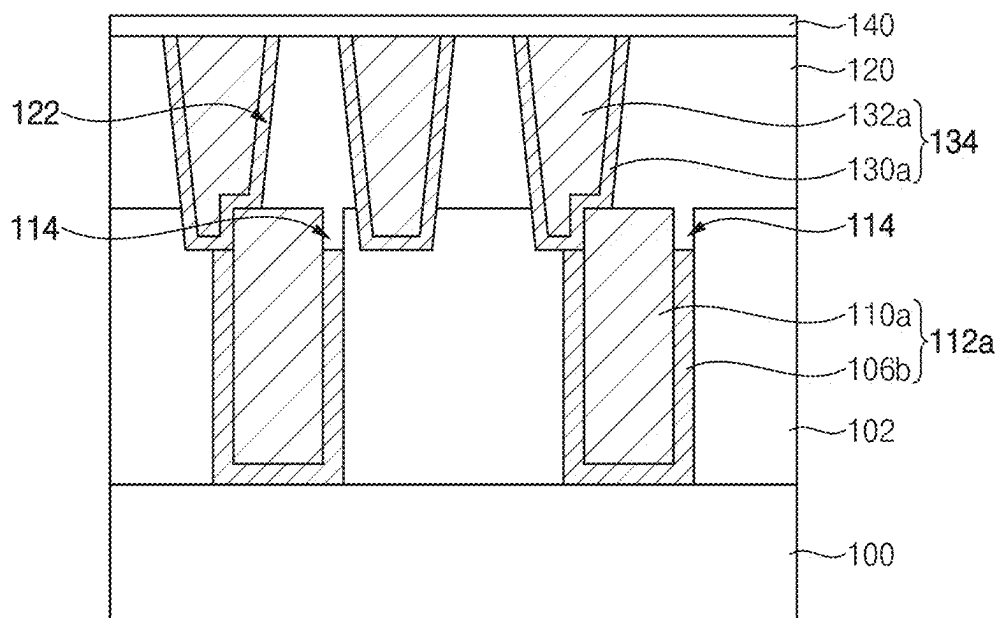


FIG. 16

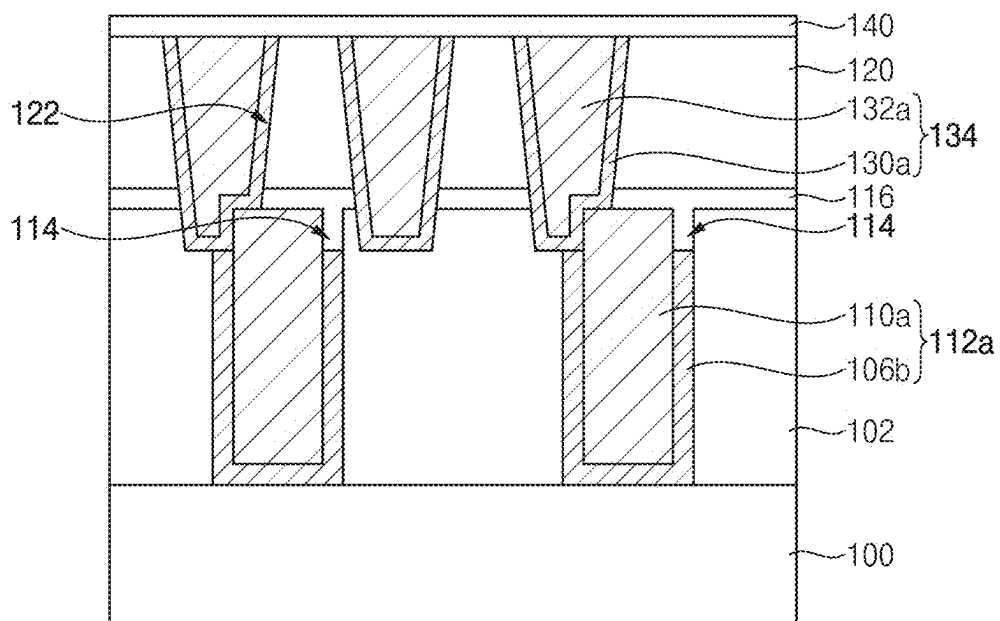


FIG. 17

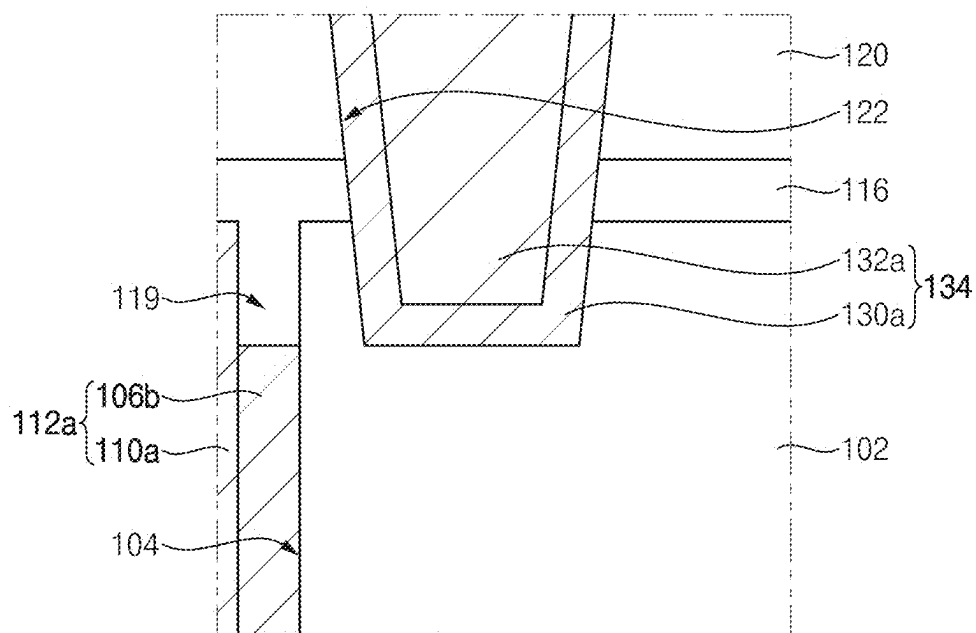


FIG. 18

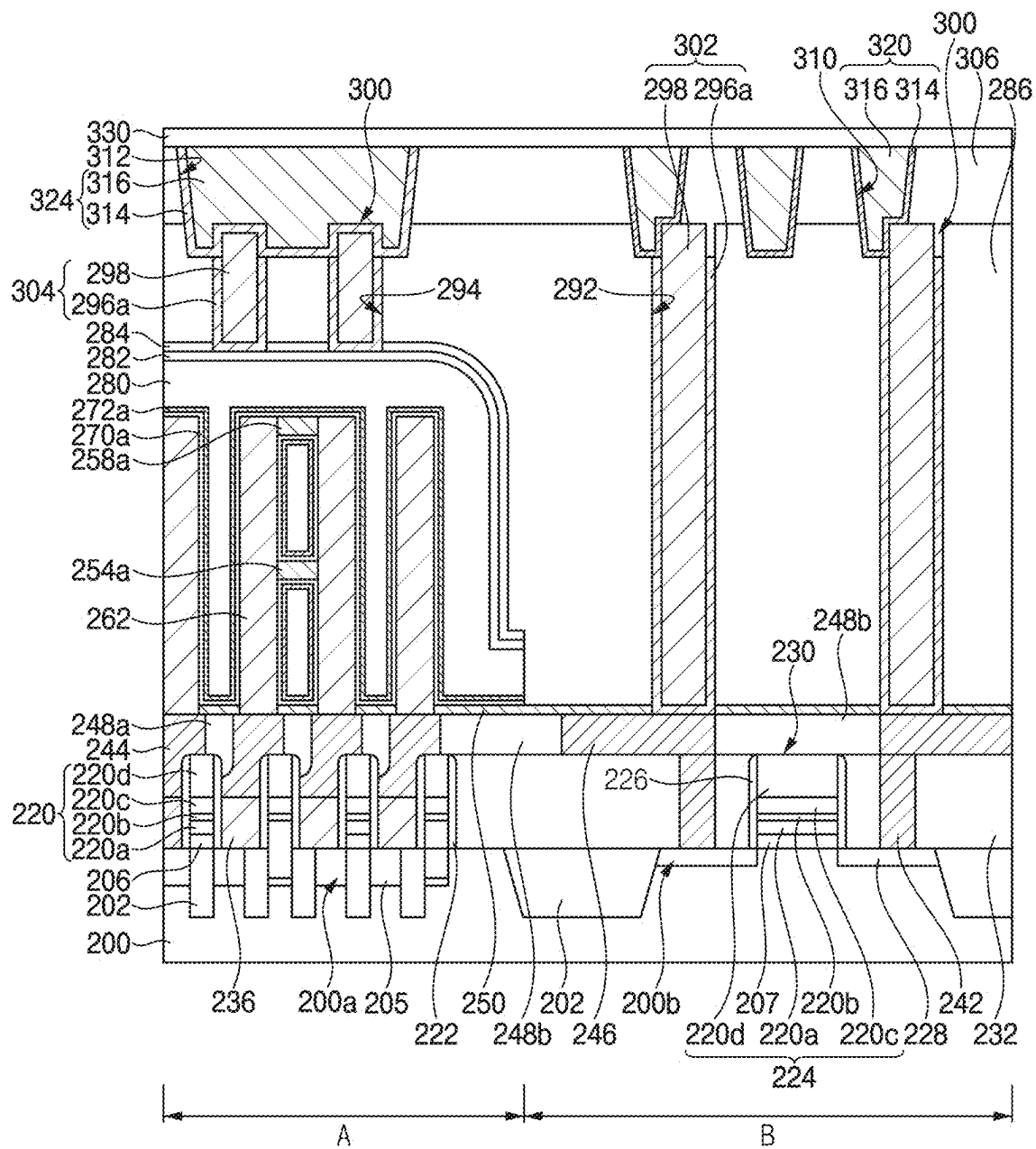


FIG. 19

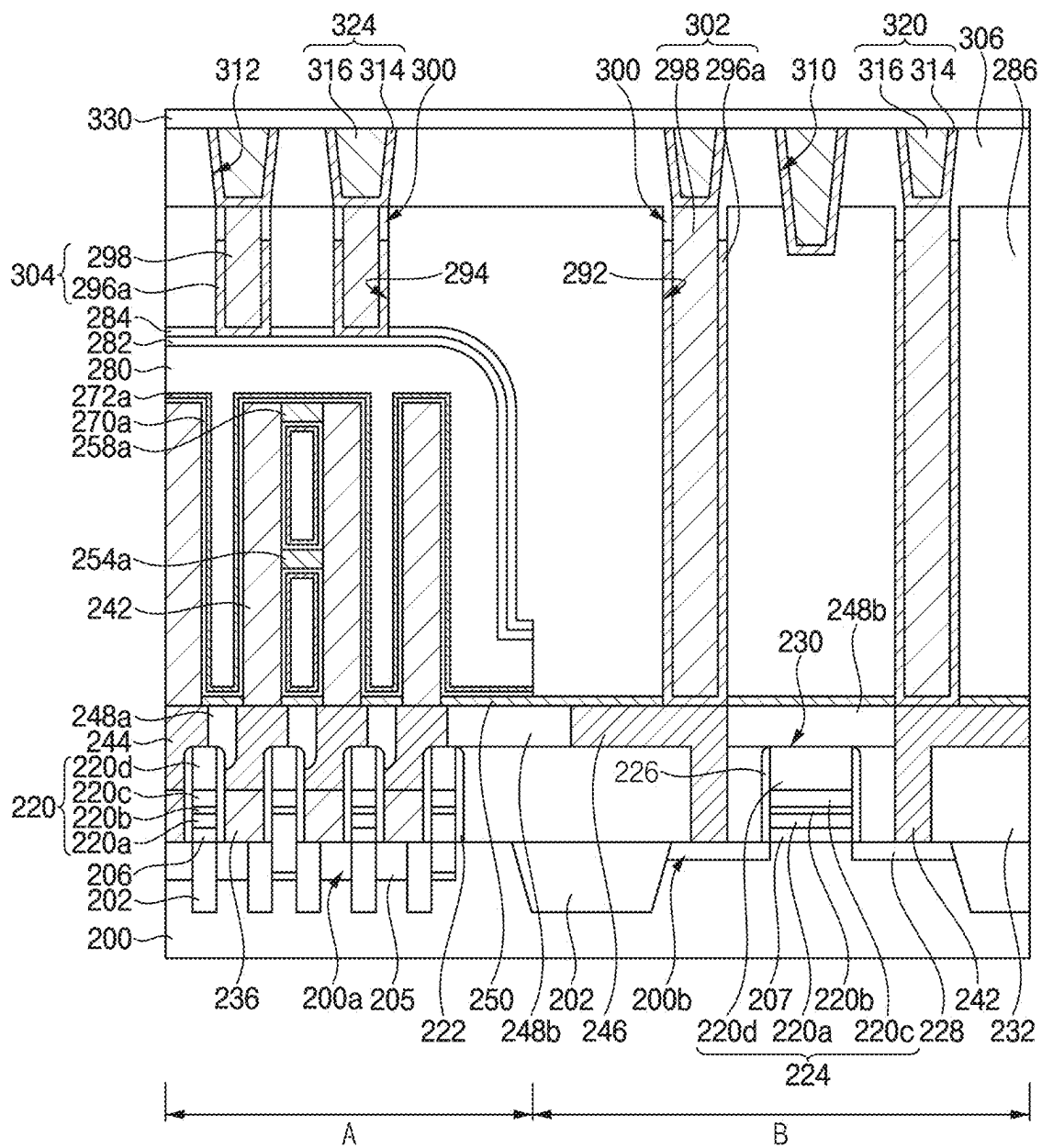


FIG. 20

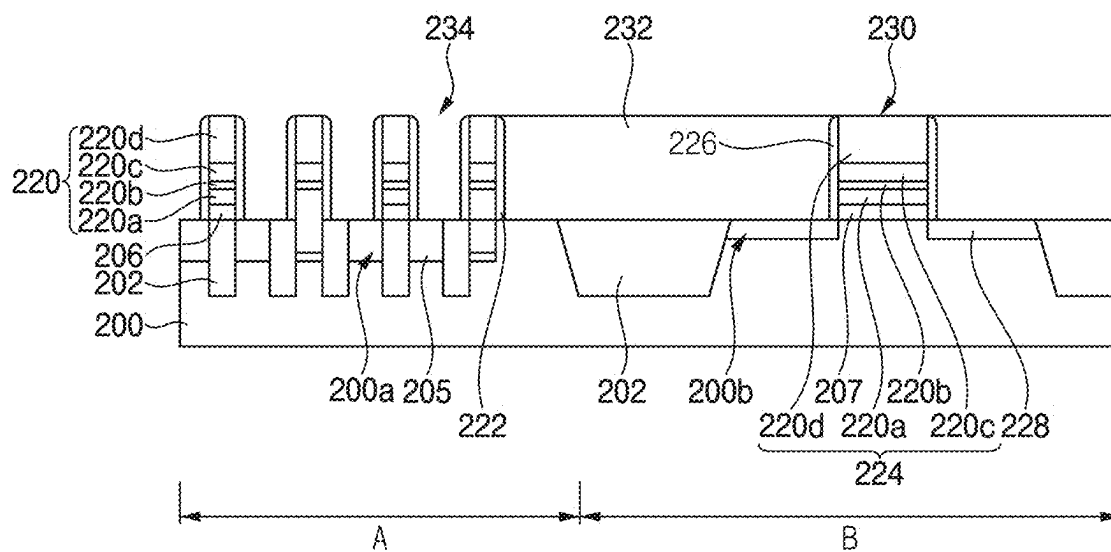


FIG. 21

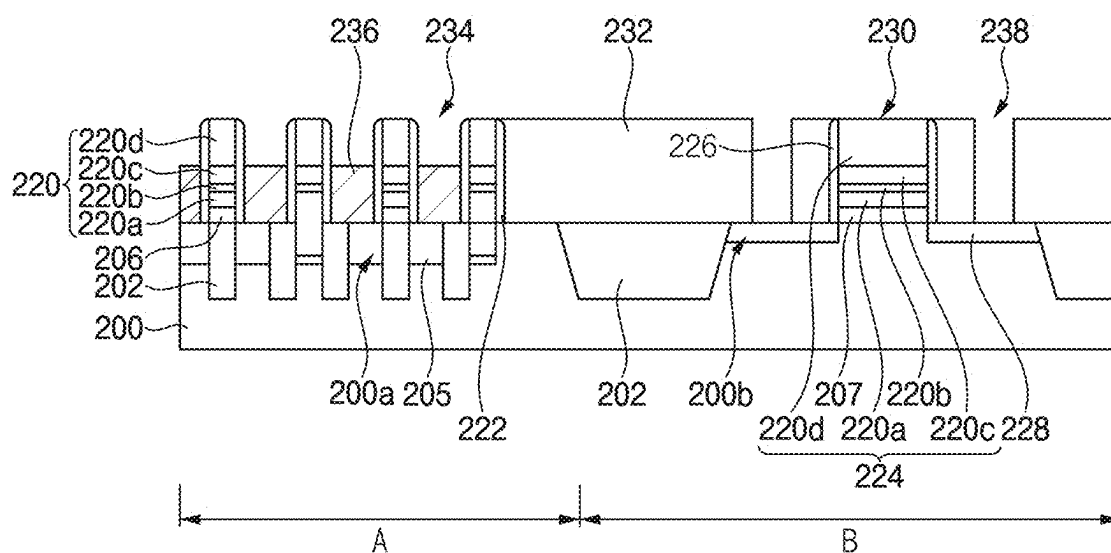


FIG. 24

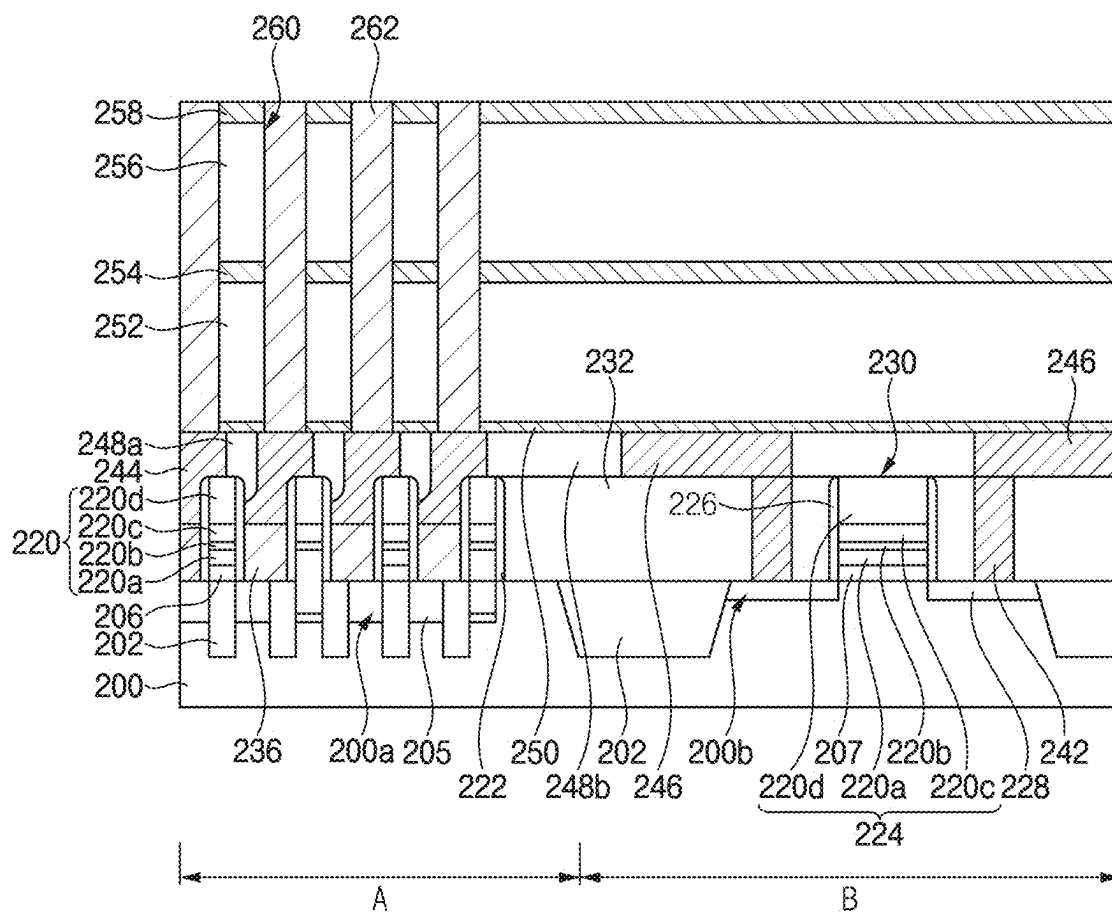


FIG. 25

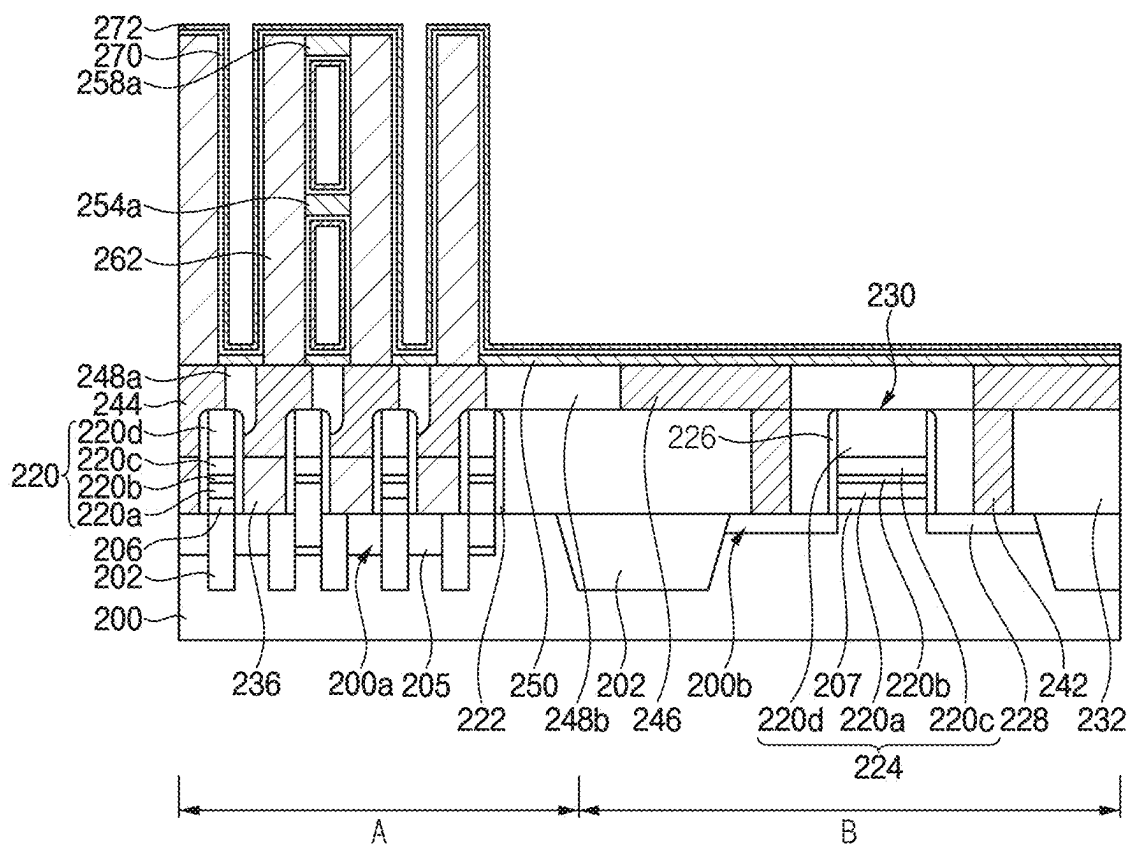


FIG. 26

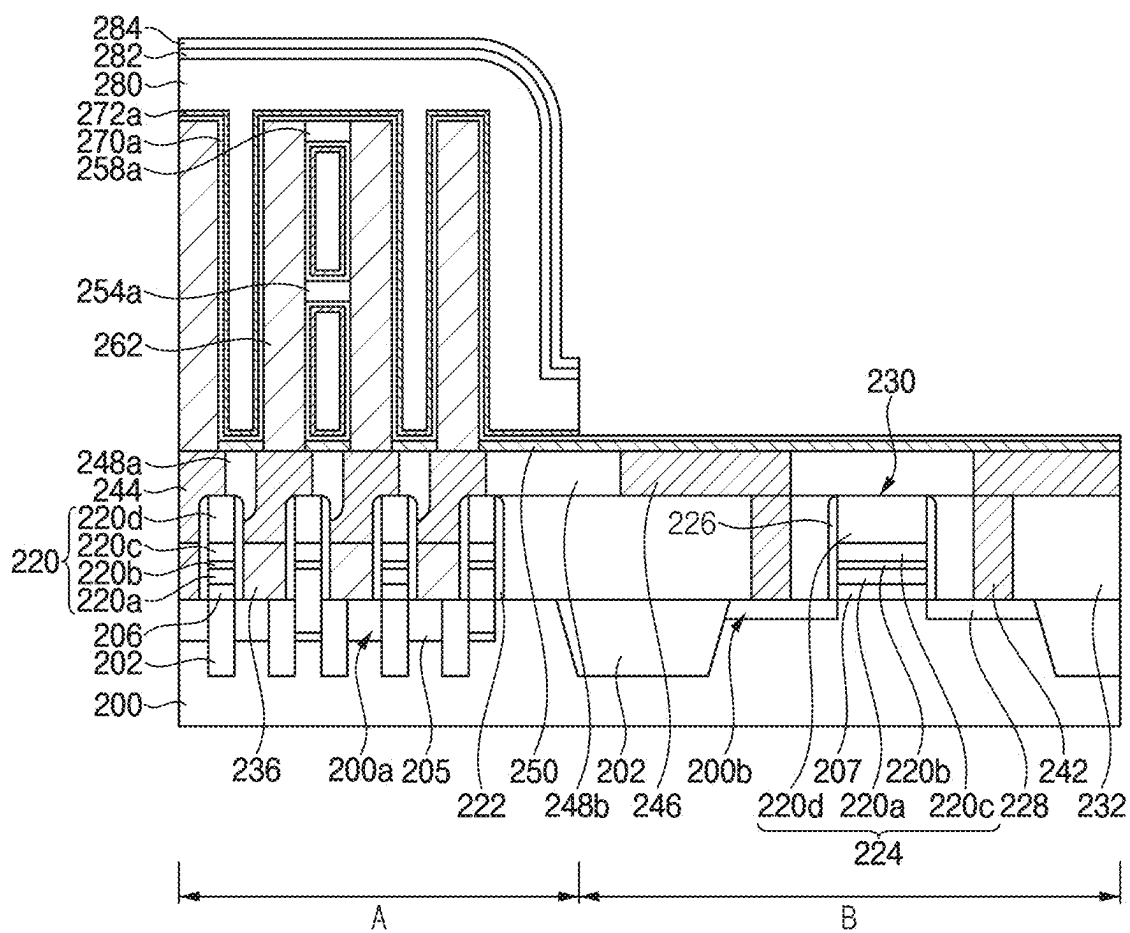


FIG. 27

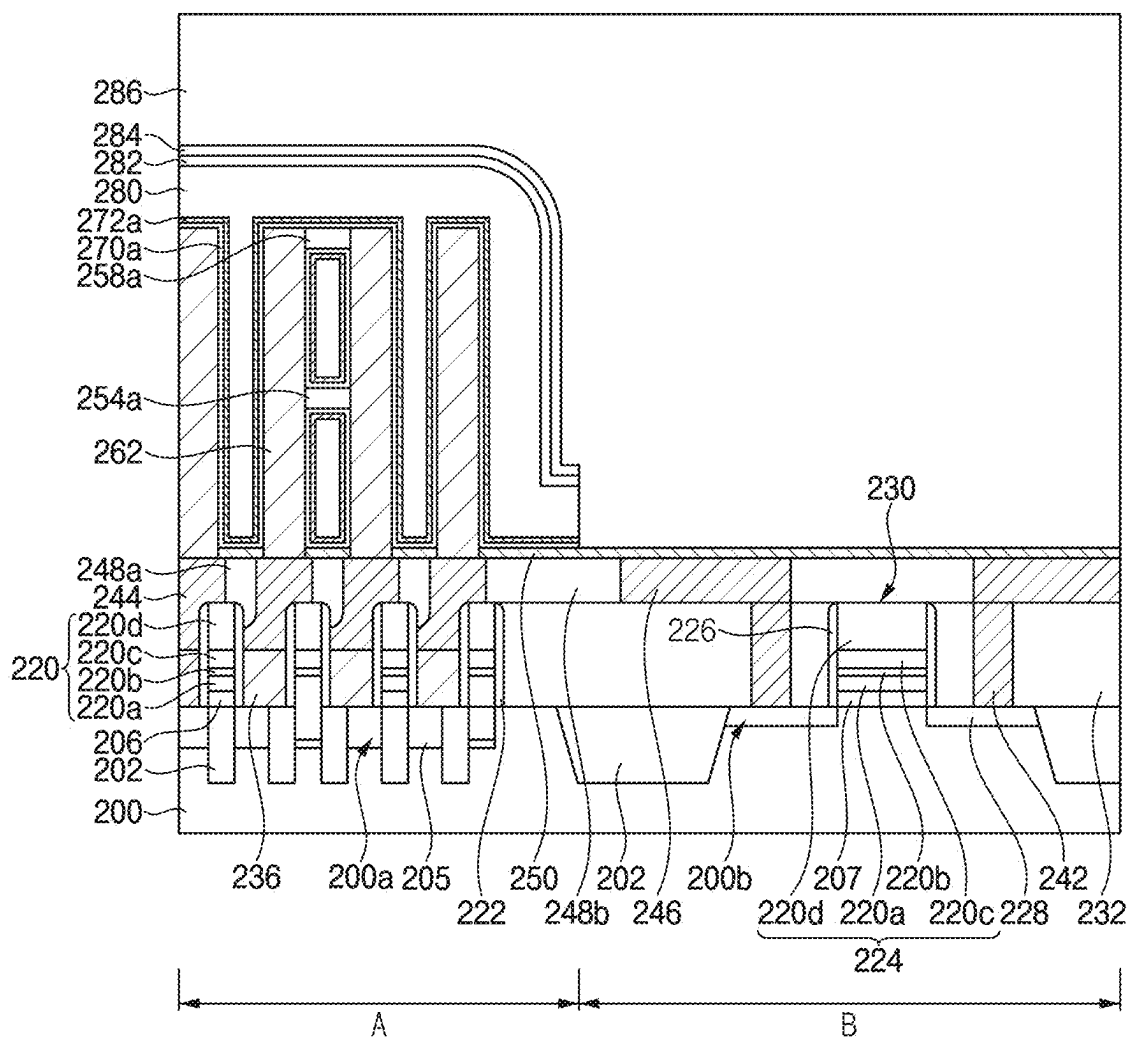


FIG. 29

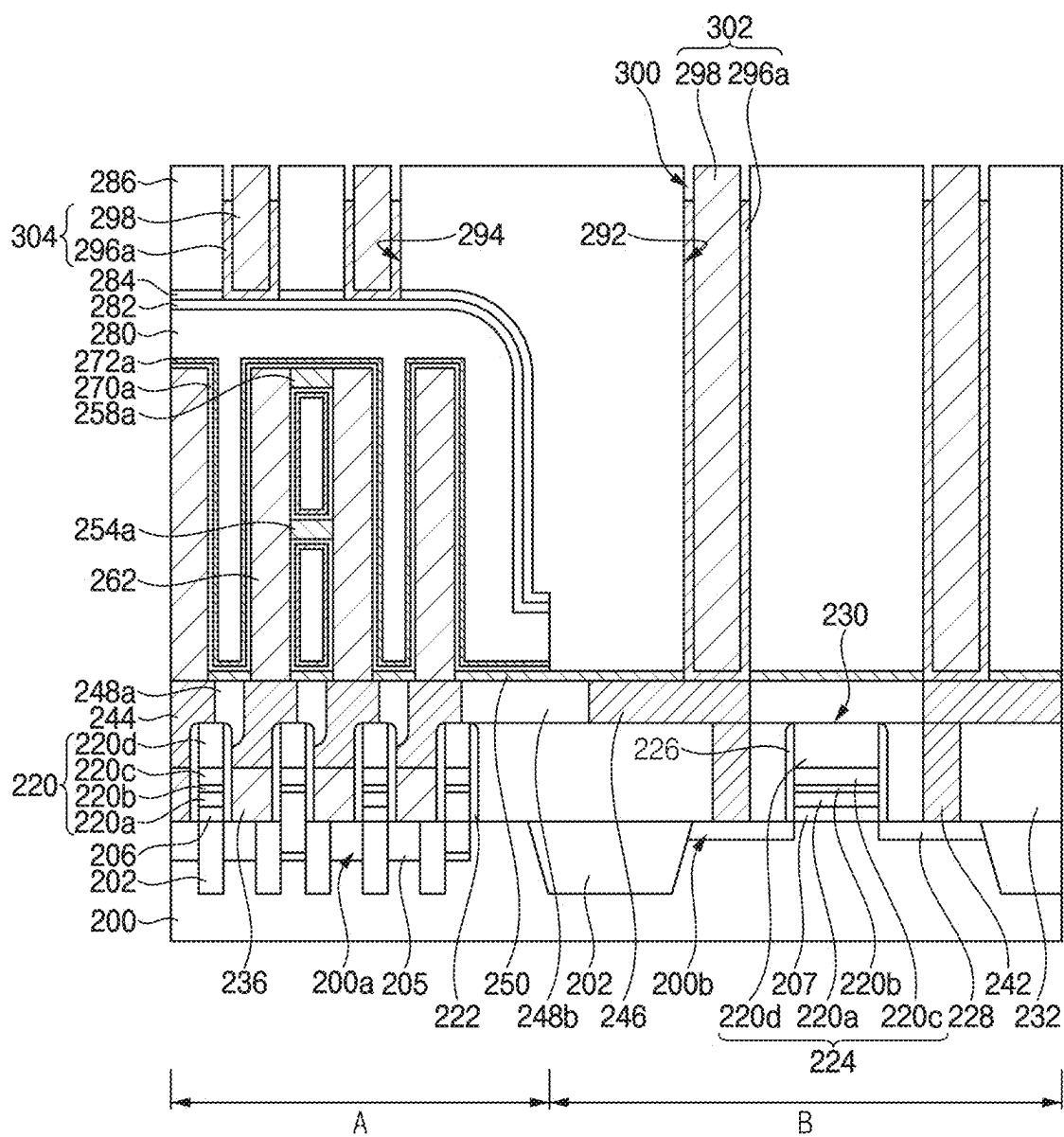


FIG. 30

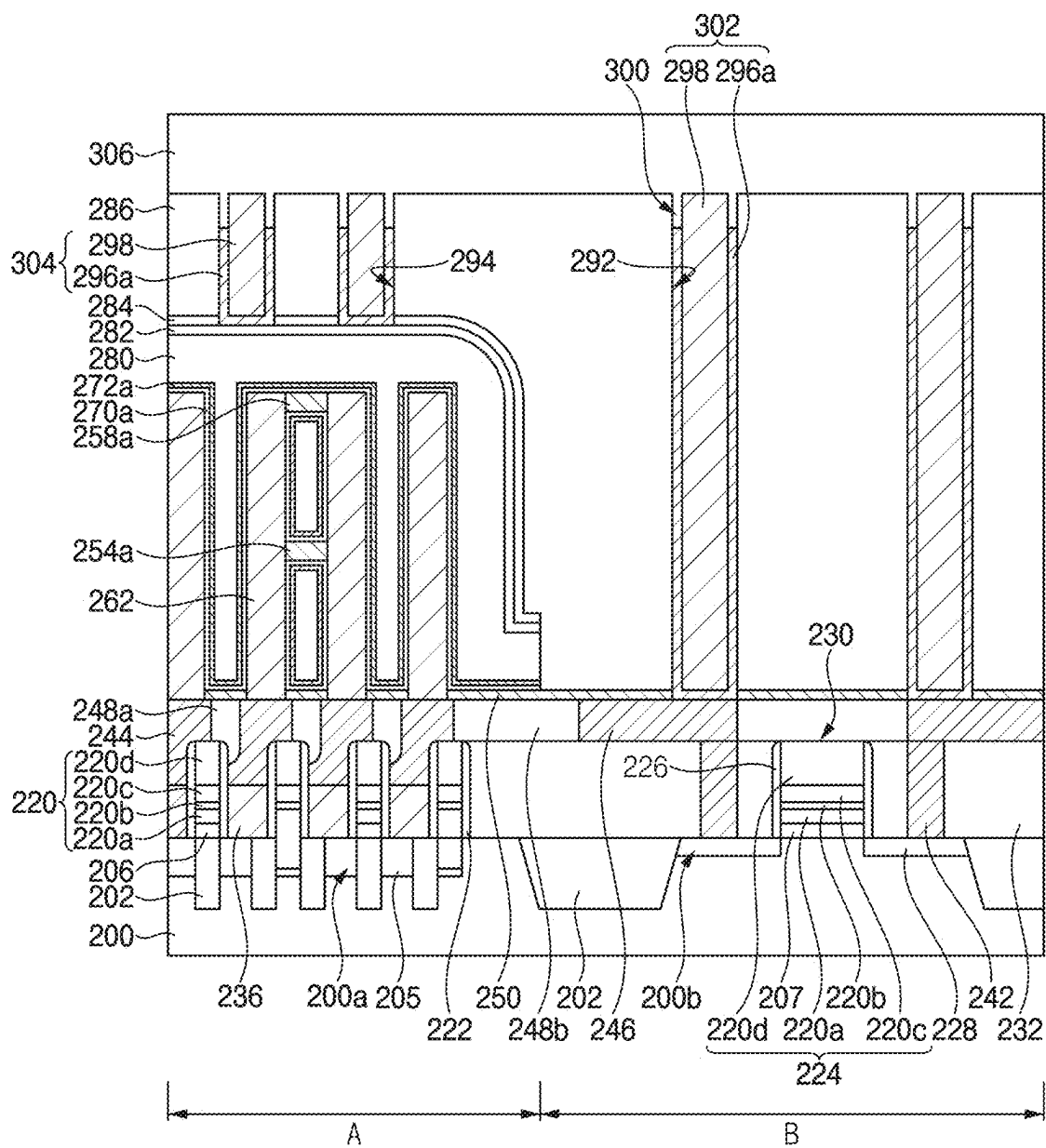


FIG. 31

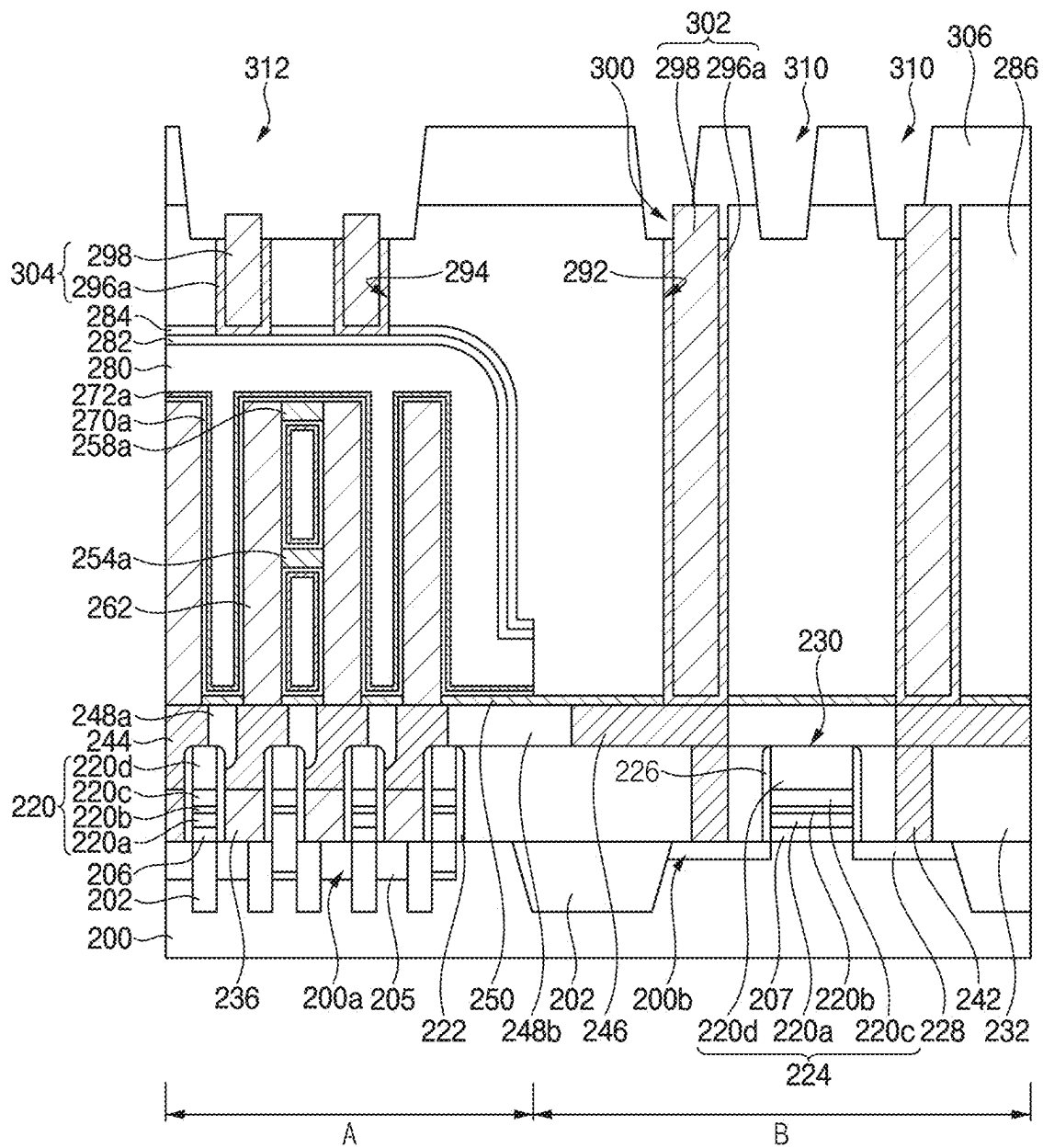


FIG. 32

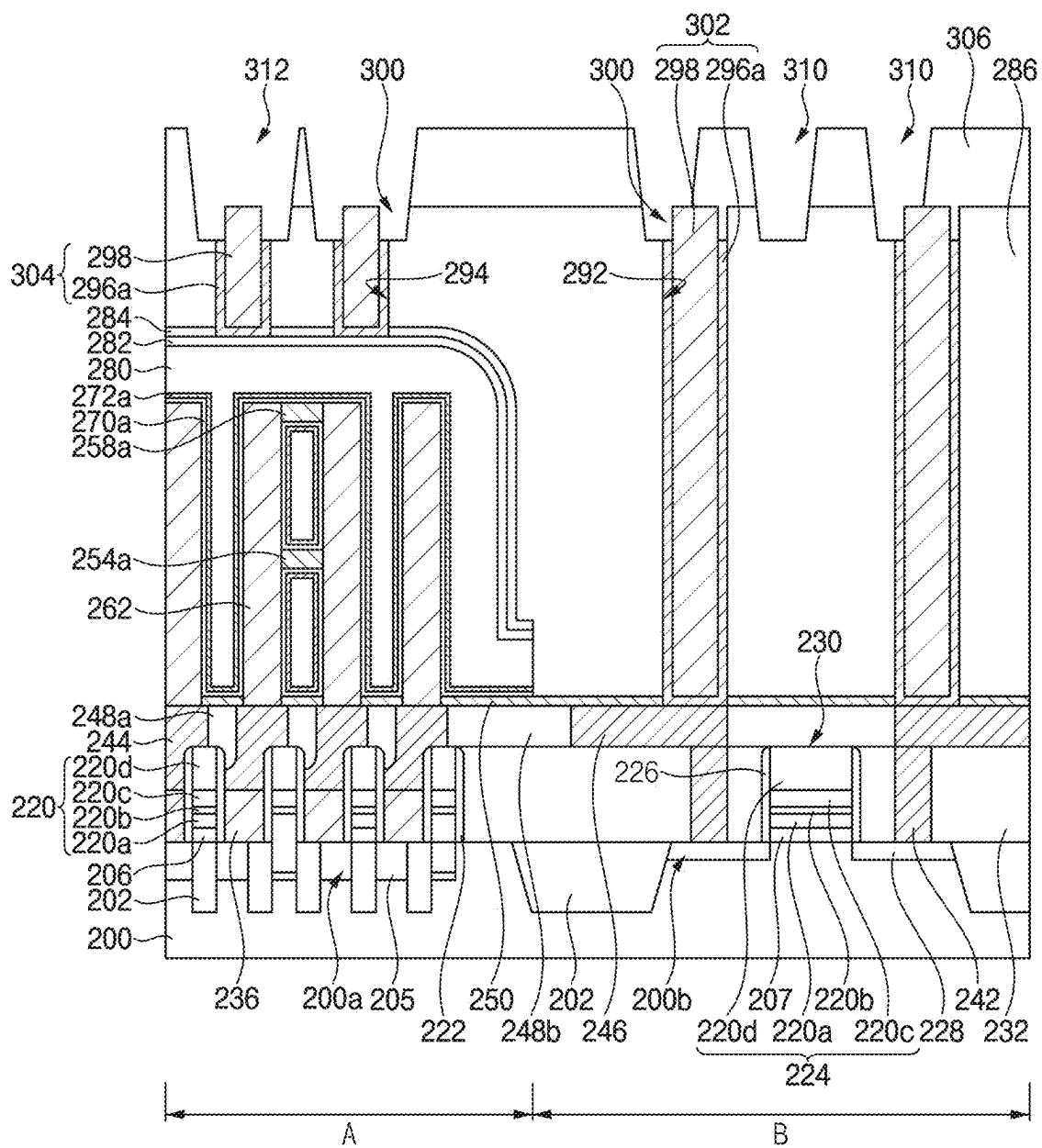


FIG. 33

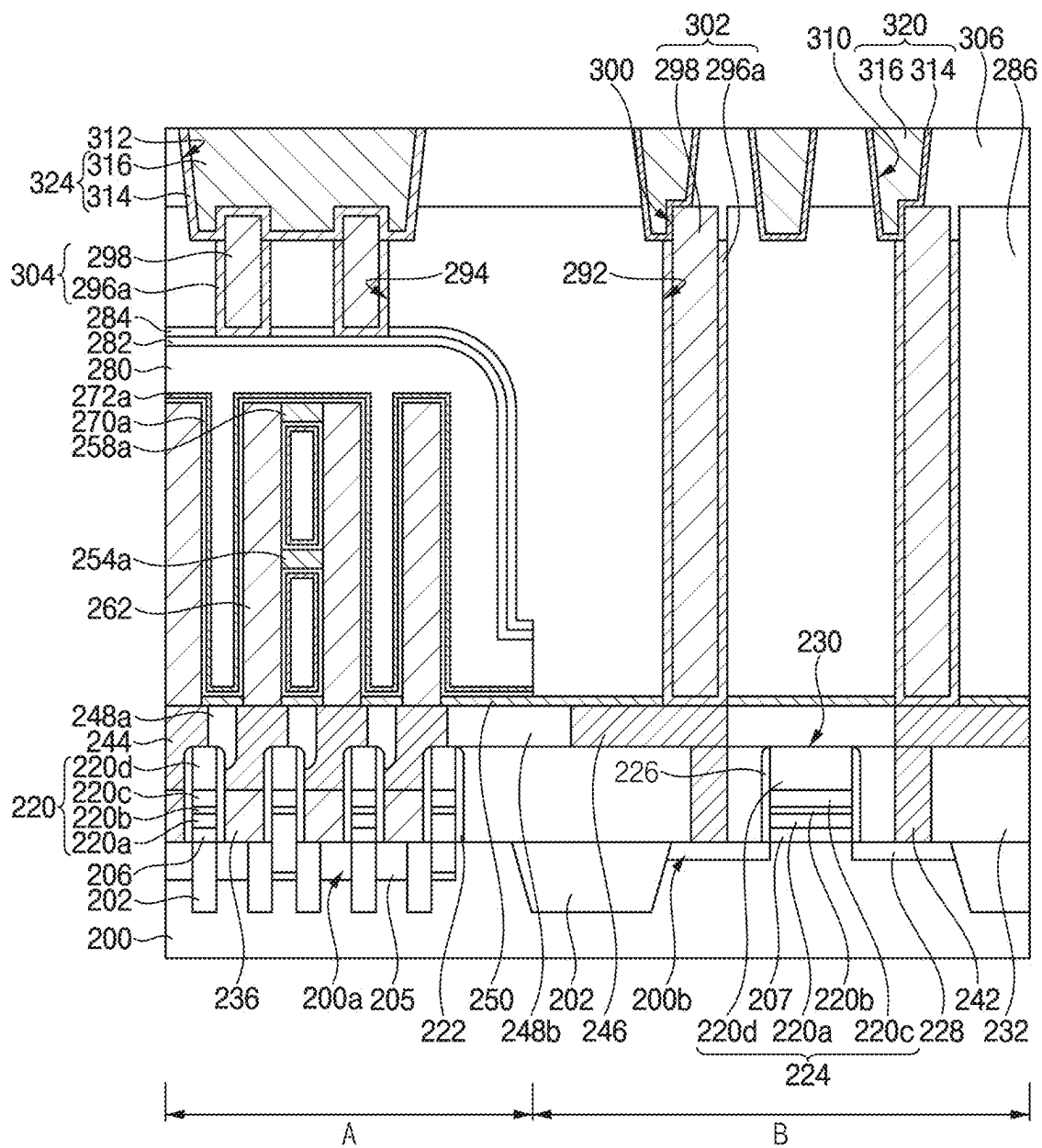


FIG. 34

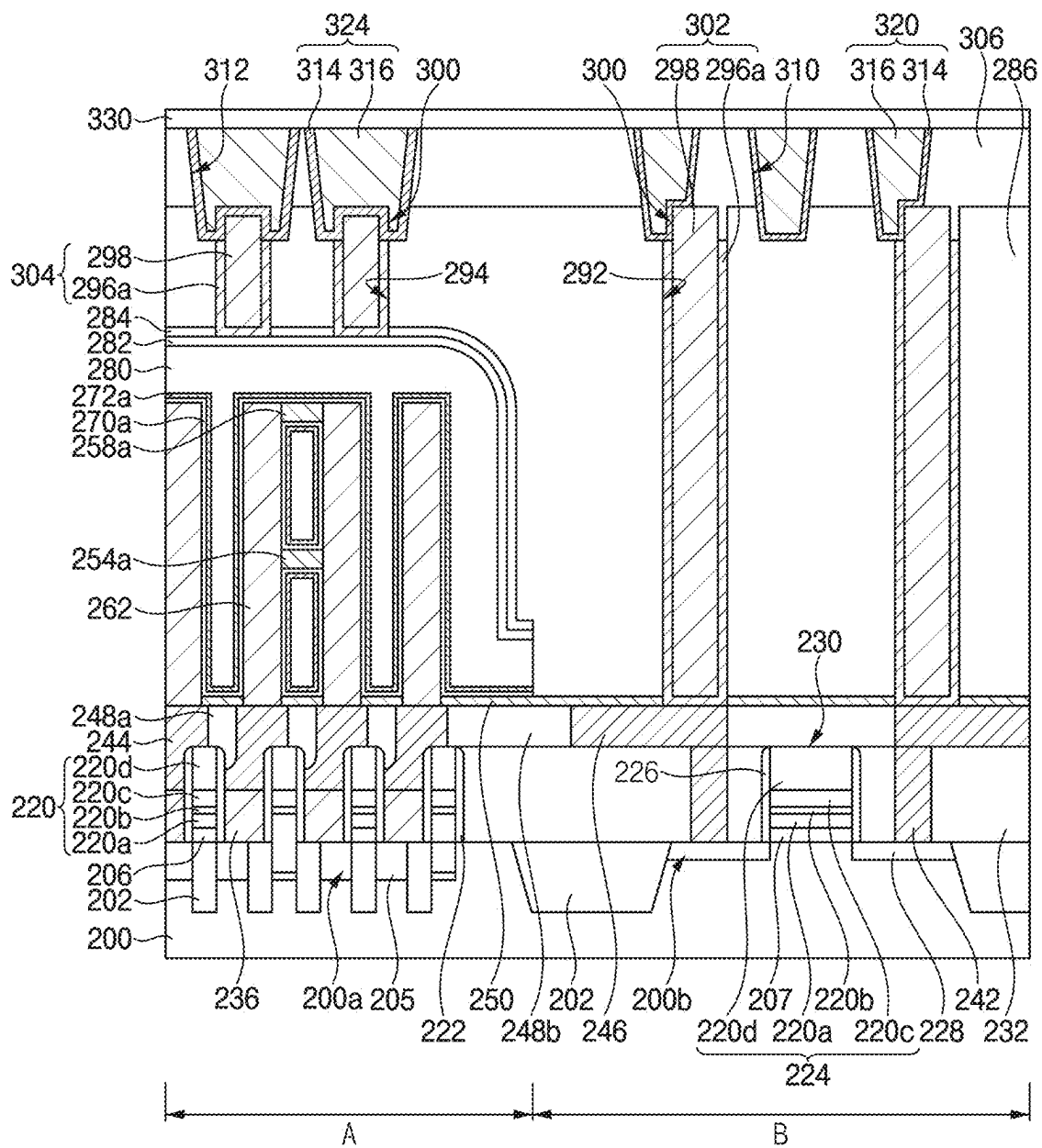
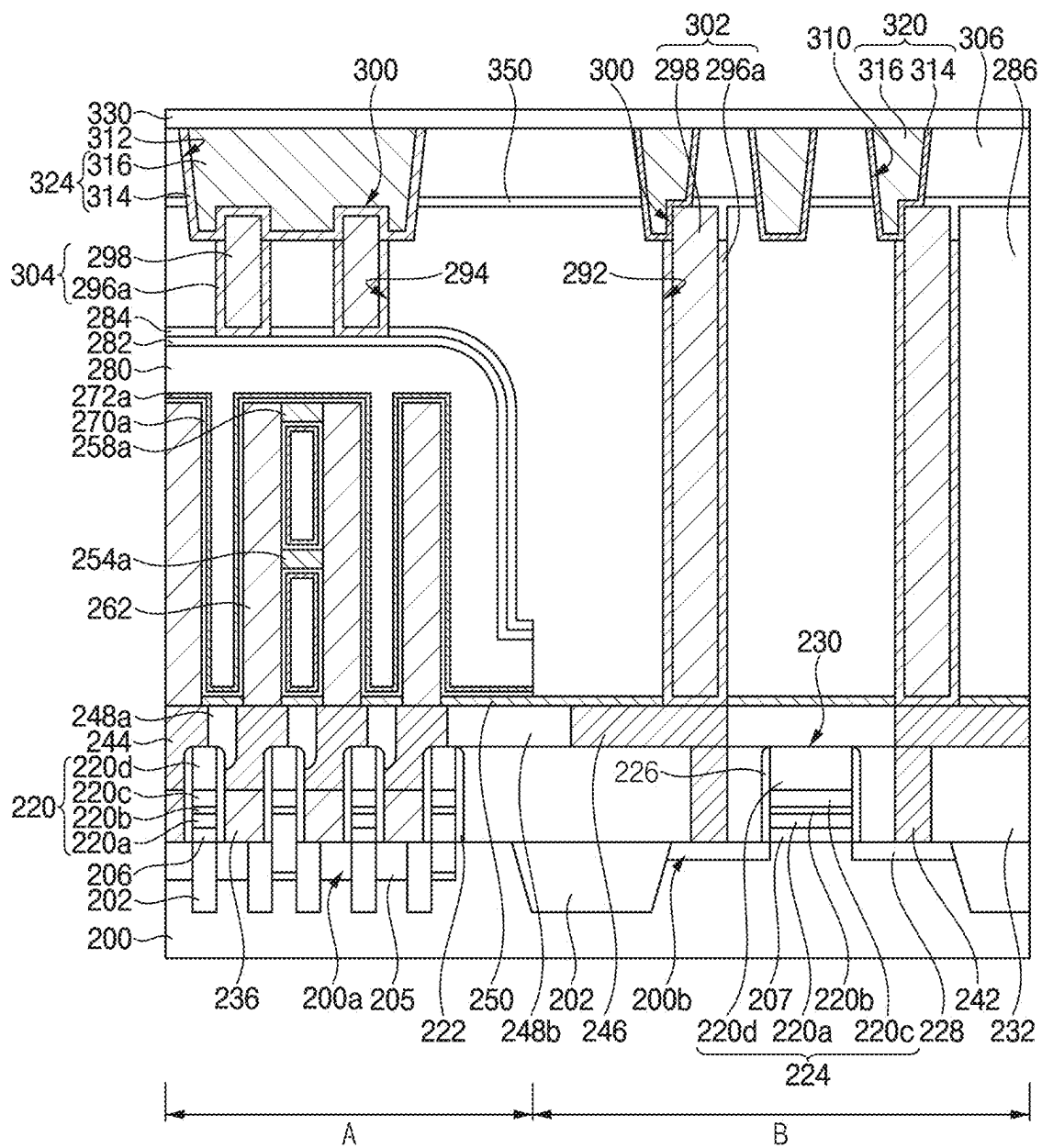


FIG. 35



SEMICONDUCTOR DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority under 35 USC § 119(a) to Korean Patent Application No. 10-2024-0033144, filed on Mar. 8, 2024, in the Korean Intellectual Property Office (KIPO), the contents of which are incorporated by reference herein in their entirety.

BACKGROUND

[0002] Various example embodiments relate to a semiconductor device. Particularly, various example embodiments relate to a semiconductor device having high reliability.

[0003] Interconnection structures for electrically connecting circuit patterns to each other may be included in a semiconductor device. An interconnection structure may include a contact plug and a conductive pattern on the contact plug. As physical dimensions of the contact plug and the conductive pattern are becoming smaller, less space is present between the contact plug and conductive patterns adjacent to the contact plug in a horizontal direction. Accordingly, insulation breakdown of insulation material between the contact plug and the conductive pattern may occur, so that a reliability of the semiconductor device may be decreased.

SUMMARY

[0004] Various example embodiments provide a semiconductor device including an interconnection structure having high reliability.

[0005] According to some example embodiments, there is provided a semiconductor device. The semiconductor device may include a substrate, a first insulating interlayer on the substrate, a plurality of first contact plugs passing through the first insulating interlayer, a second insulating interlayer that is on the first insulating interlayer and is further on the plurality of first contact plugs, and a plurality of first conductive patterns passing through the second insulating interlayer and extending below an upper surface of the first insulating interlayer in a vertical direction. Each of the plurality of first contact plugs may include a first metal pattern and a first barrier metal pattern covering a portion of a sidewall of the first metal pattern and further covering a bottom of the first metal pattern, and the first barrier metal pattern further exposing an upper sidewall of the first metal pattern. Each of the plurality of first conductive patterns may have a first bottom surface contacting a portion of an upper surface of a respective first contact plug of the plurality of first contact plugs, and a second bottom surface contacting the first insulating interlayer and being lower than the first bottom surface.

[0006] According to some example embodiments, there is provided a semiconductor device. The semiconductor device may include a substrate, a first insulating interlayer on the substrate, a plurality of first contact plugs, a second insulating interlayer that is on the first insulating interlayer and is further on the plurality of first contact plugs, and a plurality of first conductive patterns passing through the second insulating interlayer and extending below an upper surface of the first insulating interlayer in a vertical direction. The first insulating interlayer may include a plurality of contact holes. The plurality of first contact plugs may be in

the plurality of contact holes, respectively. Each of the plurality of first contact plugs may include a first metal pattern and a first barrier metal pattern covering a portion of a sidewall of the first metal pattern and further covering a bottom of the first metal pattern. An uppermost surface of the first barrier metal pattern may be lower than an upper surface of the first metal pattern. Each of the plurality of first conductive patterns may contact an upper surface of a respective first contact plug of the plurality of first contact plugs and may further contact a respective upper portion of the first insulating interlayer. A distance in a horizontal direction between an uppermost sidewall of the first metal pattern of the respective first contact plug and the first conductive pattern may be greater than a distance in the horizontal direction between an uppermost sidewall of the first barrier metal pattern of the respective first contact plug and the first conductive pattern.

[0007] According to some example embodiments, there is provided a semiconductor device. The semiconductor device may include a substrate, a plurality of capacitors on a cell region of the substrate, a plurality of peripheral circuit patterns on a peripheral circuit region of the substrate, a first insulating interlayer covering the plurality of capacitors and further covering the plurality of peripheral circuit patterns, a plurality of first contact plugs passing through the first insulating interlayer and connecting to the plurality of peripheral circuit patterns, a second insulating interlayer that is on the first insulating interlayer and is further on the plurality of first contact plugs, and a first conductive pattern passing through the second insulating interlayer and extending below an upper surface of the first insulating interlayer in a vertical direction. Each of the plurality of first contact plugs may include a first metal pattern and a first barrier metal pattern covering a portion of a sidewall of the first metal pattern and further covering a bottom of the first metal pattern and the first barrier metal pattern further exposing an upper sidewall of the first metal pattern. The first conductive pattern may contact upper surfaces of the plurality of first contact plugs and further contact an upper surface of the first insulating interlayer on the peripheral circuit region. A distance in a horizontal direction between an uppermost sidewall of the first metal pattern of a first contact plug of the plurality of first contact plugs and the first conductive pattern may be greater than a distance in the horizontal direction between an uppermost sidewall of the first barrier metal pattern of the first contact plug and the first conductive pattern.

[0008] According to example embodiments, in the interconnection structure, a minimum distance in a horizontal direction between the contact plug and the conductive pattern adjacent to the contact plug may be increased. Accordingly, insulation breakdown between the contact plug and the conductive pattern may be decreased, so that the interconnection structure may have high reliability.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Various example embodiments will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings. FIGS. 1A to 35 represent various non-limiting, example embodiments as described herein.

[0010] FIG. 1A is a cross-sectional view of an interconnection structure of a semiconductor device according to example embodiments;

[0011] FIG. 1B is a cross-sectional view of an interconnection structure of a semiconductor device according to example embodiments;

[0012] FIG. 1C is a cross-sectional view of an interconnection structure of a semiconductor device according to example embodiments;

[0013] FIG. 2 is a plan view of an interconnection structure of a semiconductor device according to example embodiments;

[0014] FIG. 3 is an enlarged cross-sectional view of a portion (i.e. portion C) of FIG. 1A;

[0015] FIG. 4 is an enlarged cross-sectional view of a portion of an interconnection structure of a semiconductor device according to some example embodiments;

[0016] FIG. 5 is an enlarged cross-sectional view of a portion of an interconnection structure of a semiconductor device according to some example embodiments;

[0017] FIGS. 6 to 15 are cross-sectional views illustrating a method of manufacturing an interconnection structure of a semiconductor device according to example embodiments;

[0018] FIG. 16 is a cross-sectional view of an interconnection structure of a semiconductor device according to example embodiments;

[0019] FIG. 17 is an enlarged cross-sectional view of a portion of FIG. 16;

[0020] FIG. 18 is a cross-sectional view of a semiconductor device according to example embodiments;

[0021] FIG. 19 is a cross-sectional view of a semiconductor device according to some example embodiments;

[0022] FIGS. 20 to 34 are cross-sectional views illustrating a method of manufacturing a semiconductor device according to example embodiments; and

[0023] FIG. 35 is a cross-sectional view of a semiconductor device according to example embodiments.

DETAILED DESCRIPTION

[0024] Hereinafter, various example embodiments will be described in detail with reference to the accompanying drawings.

[0025] FIG. 1A is a cross-sectional view of an interconnection structure of a semiconductor device according to example embodiments. FIG. 1B is a cross-sectional view of an interconnection structure of a semiconductor device according to example embodiments. FIG. 1C is a cross-sectional view of an interconnection structure of a semiconductor device according to example embodiments. FIG. 2 is a plan view of an interconnection structure of a semiconductor device according to example embodiments. FIG. 3 is an enlarged cross-sectional view of a portion (i.e. portion C) of FIG. 1A. FIG. 4 is an enlarged cross-sectional view of a portion of an interconnection structure of a semiconductor device according to some example embodiments. FIG. 5 is an enlarged cross-sectional view of a portion of an interconnection structure of a semiconductor device according to some example embodiments.

[0026] Each of FIGS. 1A, 1B, and 1C is a cross-sectional view taken along line I-I' of FIG. 2. The interconnection structures shown in FIGS. 4 and 5 may be substantially the same as the interconnection structure shown in FIG. 1A, except for materials occupying a first recess or a shape of the materials occupying the first recess.

[0027] Referring to FIGS. 1A, 1B, 1C, 2, and 3, a first insulating interlayer 102 may be on a substrate 100. In example embodiments, the first insulating interlayer 102

may include a silicon oxide-based material. For example, the first insulating interlayer 102 may include Tetra Ethyl Ortho Silicate (TEOS).

[0028] First contact holes 104 may pass through the first insulating interlayer 102, and may expose a surface of the substrate 100. In example embodiments, the surface of the substrate 100 may be exposed by bottoms of the first contact holes 104. In some example embodiments, a lower conductive structure may be further on the substrate 100, and the lower conductive layer may be exposed by bottoms of the first contact holes 104.

[0029] In example embodiments, in a plan view, each of the first contact holes 104 may have a circular shape or an oval shape.

[0030] A first contact plug 112a may be included for each of the first contact holes 104, each first contact plug 112b being in a respective one of the first contact holes 104. The first contact plugs 112a may pass through the first insulating interlayer 102. In example embodiments, bottoms of the first contact plugs 112a may contact the surface of the substrate 100. In some example embodiments in which a lower conductive structure is further on the substrate 100, the bottoms of the first contact plugs 112a may contact the lower conductive structure.

[0031] The first contact plugs 112a may each include a first barrier metal pattern 106b and a first metal pattern 110a. The first barrier metal pattern 106b may cover a portion of a sidewall of the first metal pattern 110a and further cover a bottom of the first metal pattern 110a.

[0032] The first barrier metal pattern 106b may conform to the portion of the sidewall of the first metal pattern 110a and further to the bottom of each of the first contact holes 104. In example embodiments, the first barrier metal pattern 106b on the portion of the sidewall of the first metal pattern 110a and on the bottom of each of the first contact holes 104 may have a uniform thickness. The first metal pattern 110a may be on the first barrier metal pattern 106b to occupy the first contact hole 104. Accordingly, the first barrier metal pattern 106b may surround the portion of the sidewall of the first metal pattern 110a and may further surround the bottom of the first metal pattern 110a.

[0033] An upper surface of the first metal pattern 110a may be substantially coplanar with an upper surface of the first insulating interlayer 102. An uppermost surface of the first barrier metal pattern 106b may be lower than the upper surface of the first metal pattern 110a. The first barrier metal pattern 106b may not be on an upper sidewall of the first metal pattern 110a. An upper sidewall of the first metal pattern 110a may be exposed by the first barrier metal pattern 106b. The first barrier metal pattern 106b may not be disposed between the upper sidewall of the first metal pattern 110a and the first insulating interlayer 102.

[0034] A first recess 114 may be between the upper sidewall of the first metal pattern 110a and the first insulating interlayer 102, and the first recess 114 may surround the upper sidewall of the first metal pattern 110a. The first recess 114 may be disposed above the uppermost surface of the first barrier metal pattern 106b.

[0035] In example embodiments, a sidewall of the first contact hole 104 may have a vertical slope, or may have a sidewall slope such that an inner width of the first contact hole 104 decreases from upper portion to the bottom thereof. Accordingly, a sidewall of the first contact plug 112a may have the vertical slope, or may have a sidewall slope such

that a width of the first contact plug **112a** decreases from upper portion to the bottom thereof.

[0036] In example embodiments, the first barrier metal pattern **106b** may include, e.g., titanium, titanium nitride, tantalum, tantalum nitride, or tungsten nitride.

[0037] In example embodiments, the first metal pattern **110a** may include tungsten. In some example embodiments, the first metal pattern **110a** may include aluminum, copper, ruthenium, cobalt, nickel, palladium, platinum, silver, or gold.

[0038] A second insulating interlayer **120** may occupy the first recess **114**, and may cover upper surfaces of the first contact plugs **112a** and of the first insulating interlayer **102**. The second insulating interlayer **120** may include a silicon oxide-based material. The second insulating interlayer **120** may include a material having a dielectric constant lower than a dielectric constant of the first insulating interlayer **102**. The second insulating interlayer **120** may include a low-K insulation material. For example, the second insulating interlayer **120** may include SiOC or SiOF. The same insulation material (e.g., the second insulating interlayer) may be in the first recess **114**, on the first insulating interlayer **102**, and on the first contact plug **112a**. For example, the first recess **114** may not be occupied with nitride, such as silicon nitride.

[0039] In example embodiments, as shown in FIGS. 1A, 1B, 1C, and 3, voids may not be included in the second insulating interlayer **120** inside the first recess **114**. In some example embodiments, as shown in FIG. 4, a void **118** may be included in the second insulating interlayer **120** inside the first recess **114**.

[0040] In some example embodiments, as shown in FIG. 5, the first recess **114** may not be occupied with the second insulating interlayer **120** therein, so that the first recess **114** may remain as a space including, e.g., air **119**. In this case, the second insulating interlayer **120** may be disposed on the first insulating interlayer **102** and on the first contact plug **112a** to cover an inlet portion of the first recess **114**.

[0041] Openings **122** may pass through the second insulating interlayer **120**, and may extend below the upper surface of the first insulating interlayer **102** in a vertical direction (i.e., a direction perpendicular to the upper surface of the substrate **100**). The first contact plug **112a** may be exposed by at least a lower portion of each of the openings **122**. Each of the openings **122** may have a sidewall slope such that an inner width of each of the openings **122** decreases from an upper portion to a bottom thereof. Each of the openings **122** may have a lower width that is less than an upper width.

[0042] A first conductive pattern **134** may be in each of the openings **122**. The first conductive pattern **134** may pass through the second insulating interlayer **120**, and may extend below the upper surface of the first insulating interlayer **102**.

[0043] A first bottom portion of the first conductive pattern **134** may contact the upper surface of the first contact plug **112a**. A second bottom portion of the first conductive pattern **134** may contact the first insulating interlayer **102**.

[0044] In example embodiments, the first conductive pattern **134** may not entirely cover the upper surface of a first contact plug **112a**, but may partially cover the upper surface of a first contact plug **112a**. In example embodiments, a lower portion of the first conductive pattern **134** may contact the upper surface of the first metal pattern **110a** and further

contact an upper sidewall of the first metal pattern **110a**, included in the first contact plug **112a**, and the lower portion of the first conductive pattern **134** may further contact an upper surface of the first barrier metal pattern **106b**. In example embodiments, a portion of the first conductive pattern **134** may be disposed in a region corresponding to but opposite from the first recess **114** relative to, e.g., the first contact plug **112a**. The first bottom portion of the first conductive pattern **134** may be higher than the second bottom portion of the first conductive pattern **134**. In example embodiments, the first bottom portion of the first conductive pattern **134** may be substantially coplanar with the upper surface of the first metal pattern **110a** of the first contact plug **112a**. The second bottom portion of the first conductive pattern **134** may be lower than the upper surface of the first insulating interlayer **102**, and the second bottom portion further may be higher than a bottom of the first insulating interlayer **102**. The second bottom portion of the first conductive pattern **134** may be at an inner portion of the first insulating interlayer **102**.

[0045] In example embodiments, as shown in FIGS. 1A and 1C, the second bottom portion of the first conductive pattern **134** may be substantially coplanar with a bottom of the first recess **114**. The second bottom portion of the first conductive pattern **134** may be substantially coplanar with the uppermost surface of the first barrier metal pattern **106b**. In some example embodiments, as shown in FIG. 1B, the second bottom portion of the first conductive pattern **134** may be lower than the bottom of the first recess **114**. The second bottom portion of the first conductive pattern **134** may be lower than the uppermost surface of the first barrier metal pattern **106b**.

[0046] The first conductive pattern **134** may include a second barrier metal pattern **130a** and a second metal pattern **132a**. The second barrier metal pattern **130a** may cover a sidewall of the second metal pattern **132a** and may further cover a bottom of the second metal pattern **132a**.

[0047] The second barrier metal pattern **130a** may conform to the sidewall of the opening **122** and to the bottom of the opening **122**. In example embodiments, the second barrier metal pattern **130a** on the sidewall and on the bottom of the opening **122** may have a uniform thickness. In some example embodiments, the second barrier metal pattern **130a** on the bottom of the opening **122** may have a thickness greater than a thickness of the second barrier metal pattern **130a** on the sidewall of the opening **122**.

[0048] The second metal pattern **132a** may be on the second barrier metal pattern **130a** to occupy the opening **122**.

[0049] Upper surfaces of the second metal pattern **132a** and of the second barrier metal pattern **130a** may be substantially coplanar with an upper surface of the second insulating interlayer **120**.

[0050] In example embodiments, the second barrier metal pattern **130a** may include titanium, titanium nitride, tantalum, tantalum nitride, tungsten nitride, or the like. The second metal pattern **132a** may include, e.g., copper. In some example embodiments, the second metal pattern **132a** may include aluminum, ruthenium, cobalt, nickel, palladium, platinum, silver, gold or the like.

[0051] In the first conductive pattern **134**, a portion lower than the upper surface of the first insulating interlayer **102** is referred to within the following disclosure as a first portion of the first conductive pattern **134**, and a portion higher than

the upper surface of the first insulating interlayer is referred to within the following disclosure as a second portion of the first conductive pattern 134. In the claims, the recitation of a “first portion” or a “second portion” without more is not intended to incorporate these definitions therein. The first portion of the first conductive pattern 134 may contact the first insulating interlayer 102 without contacting the upper surface of the first contact plug 112a.

[0052] A width of an uppermost portion of the first conductive pattern 134 may be greater than a width of a lowermost portion of the first conductive pattern 134. At least the first portion of the first conductive pattern 134 may have a sidewall slope such that the width of first conductive pattern 134 gradually decreases downward.

[0053] In example embodiments, as shown in FIGS. 1A and 1B, the first conductive pattern 134 may have the sidewall slope such that the width of the first conductive pattern 134 gradually decreases from the uppermost portion of the first conductive pattern 134 to the lowermost portion of the first conductive pattern 134. The sidewall slope of the first conductive pattern 134 may be constant. In some example embodiments, as shown in FIG. 1C, an upper sidewall of the first conductive pattern 134 may have a vertical slope, and a lower sidewall of the first conductive pattern 134 may have a slope such that the width of first conductive pattern 134 gradually decreases downward.

[0054] In example embodiments, a portion of at least one first conductive pattern contacting the first insulating interlayer 102 may be disposed between portions of two first conductive patterns 134 contacting the upper surface of the first contact plug 112a. The first portion of the first conductive pattern 134 may be disposed adjacent to the first contact plug 112a while being spaced apart from the first contact plug 112a. In a cross-sectional view, an upper sidewall of the first contact plug 112a and the first portion of the first conductive pattern 134 may face toward each other in a horizontal direction with respect to the upper surface of the substrate.

[0055] In example embodiments, an upper portion of the first portion of the first conductive pattern 134 may face the sidewall of the first metal pattern 110a, and a lower portion of the first portion of the first conductive pattern 134 may face a sidewall of the first barrier metal pattern 106b. In some example embodiments, the first portion of the first conductive pattern 134 may face only the sidewall of the first metal pattern 110a.

[0056] A distance d2 in the horizontal direction between an uppermost sidewall of the first metal pattern 110a and the first conductive pattern 134 may be greater than a distance d1 in the horizontal direction between an uppermost sidewall of the first barrier metal pattern 106b and the first conductive pattern 134.

[0057] The first and second insulating interlayers 102 and 120 may be laterally disposed between the uppermost sidewall of the first contact plug 112a and the first portion of the first conductive pattern 134.

[0058] The second insulating interlayer 120 and the first insulating interlayer 102 may be disposed in the horizontal direction between the sidewall of the first metal pattern 110a positioned higher than the uppermost surface of the first barrier metal pattern 106b and the first portion of the first conductive pattern 134. Only the first insulating interlayer

102 may be disposed between the sidewall of the first barrier metal pattern 106b and the first portion of the first conductive pattern 134.

[0059] In the following disclosure, the distance in the horizontal direction between the uppermost sidewall of the first barrier metal pattern 106b and the first conductive pattern 134 is referred to as the first distance d1. A portion having the first distance is referred to in the following disclosure as a third portion. In the claims, the recitation of a “first distance” or a “third portion” without more is not intended to incorporate these definitions therein. A third portion may be a weak region where insulation breakdown of insulation material between the first contact plug 112a and the first conductive pattern 134 easily occurs. Therefore, the weak region may compromise the reliability of the semiconductor device. In order to improve reliability of the semiconductor device, it is desirable to increase the first distance d1. Additionally, when an interface between the insulation materials (e.g., an interface between the first and second insulation interlayers) is positioned at the third portion, the insulation breakdown of the insulation material may easily occur at the interface between the insulation materials. Therefore, preferably, the interface between the insulation materials may not be positioned at the third portion.

[0060] An uppermost portion of the first portion of the first conductive pattern may have a width greater than a width of a lowermost portion of the first portion of the first conductive pattern. Therefore, if an upper portion of the first barrier metal pattern 106b is not recessed, the first distance d1 may be greatly decreased. In addition, if the upper portion of the first barrier metal pattern is not recessed, an interface between the upper surface of the first insulating interlayer and a lower surface of the second insulating interlayer may be positioned at the third portion to have a plane shape. In this case, the third portion may correspond to the interface between the upper surface of the first insulating interlayer and the lower surface of the second insulating interlayer. The interface between the first and second insulating interlayers may be may coincide with the third portion, and the first distance may be decreased. Therefore, insulation breakdown of insulation material positioned at the third position may easily occur.

[0061] However, in example embodiments, the uppermost surface of the first barrier metal pattern 106b may be lower than the upper surface of the first metal pattern 110a, and thus the first recess 144 may be included in the first contact plug 112a. In this case, the third portion may be positioned at a level lower than the uppermost surface of the first contact plug 112a. The third portion may not correspond to a portion between an uppermost portion of the first contact plug 112a and the first conductive pattern 134.

[0062] In addition, the first distance d1 in the case in which the upper portion of the first barrier metal pattern 106b is recessed may be greater than the first distance in the case in which the upper portion of the first barrier metal pattern 106b is not recessed.

[0063] The interface between the upper surface of the first insulating interlayer 102 and the lower surface of the second insulating interlayer 120 may not be positioned at the third portion. Only the first insulating interlayer 102 may be disposed at the third portion. Accordingly, the interface between the insulating interlayers positioned at the third portion may be more reliable.

[0064] The first distance $d1$ may be increased, and the interface between the first and second insulating interlayers **102** and **120** may not be positioned at the third portion. Accordingly, insulation breakdown of insulation material positioned at the third portion may be less likely.

[0065] A capping layer **140** may be disposed on the second metal pattern **132a**, the second barrier metal pattern **130a**, and the second insulating interlayer **120**. The capping layer **140** may cover the upper surfaces of the second metal pattern **132a**, the second barrier metal pattern **130a**, and the second insulating interlayer **120**. The capping layer **140** may include an insulation material, e.g., silicon nitride or silicon oxynitride.

[0066] FIGS. **6** to **15** are cross-sectional views illustrating a method of manufacturing an interconnection structure of a semiconductor device according to example embodiments.

[0067] Referring to FIG. **6**, a first insulating interlayer **102** may be formed on the substrate **100**.

[0068] The substrate **100** may include silicon, germanium, silicon-germanium, or a group III-V compound such as GaP, GaAs, or GaSb. In some example embodiments, the substrate **100** may be a Silicon On Insulator (SOI) substrate or a Germanium On Insulator (GOI) substrate.

[0069] The first insulating interlayer **102** may include a silicon oxide-based material. For example, the first insulating interlayer **102** may include Tetra Ethyl Ortho Silicate (TEOS). In example embodiments, before forming the first insulating interlayer **102**, a lower conductive structure may further be formed on an upper surface of the substrate **100**.

[0070] An etch mask pattern may be formed on the first insulating interlayer **102**, and the first insulating interlayer **102** may be etched using the etch mask pattern to form first contact holes **104**. In example embodiments, the etch mask pattern may have a structure in which an amorphous carbon layer pattern and a silicon oxynitride layer pattern are stacked. The etch mask pattern may be formed by deposition processes of the amorphous carbon layer and the silicon oxynitride layer and photolithography processes of the amorphous carbon layer and the silicon oxynitride layer.

[0071] The etching process of the first insulating interlayer **102** may include a dry etching process.

[0072] Referring to FIG. **7**, a first barrier metal layer **106** may be conformally formed on sidewalls and bottoms of the first contact holes **104** and on an upper surface of the first insulating interlayer **102**.

[0073] In example embodiments, the first barrier metal layer **106** may be formed to have a uniform thickness on the sidewalls and on the bottoms of the first contact holes **104** and the upper surface of the first insulating interlayer **102**.

[0074] The first barrier metal layer **106** may include, e.g., titanium, titanium nitride, tantalum, tantalum nitride, etc. The first barrier metal layer **106** may be formed by a chemical vapor deposition process or an atomic layer deposition process.

[0075] Referring to FIG. **8**, the first metal layer **110** may be formed on the first barrier metal layer **106** to fill the first contact holes **104**.

[0076] The first metal layer **110** may include, e.g., tungsten. In this case, the first metal layer **110** may be formed by a chemical vapor deposition process or an atomic layer deposition process using at least WF₆ as a deposition source.

[0077] In some example embodiments, the first metal layer **110** may include aluminum, copper, ruthenium, cobalt, nickel, palladium, platinum, silver, gold, etc.

[0078] Referring to FIG. **9**, the first barrier metal layer **106** and the first metal layer **110** may be planarized until the upper surface of the first insulating interlayer **102** is exposed to form a preliminary first contact plug **112** in each of the first contact holes **104**. The preliminary first contact plug **112** may include a preliminary first barrier metal pattern **106a** and a first metal pattern **110a**. The planarization process may include, e.g., a chemical mechanical polishing process and/or an etch-back process.

[0079] Referring to FIG. **10**, an upper portion of the preliminary first barrier metal pattern **106a** may be partially etched to form a first barrier metal pattern **106b**. Accordingly, the first contact plug **112a** including the first barrier metal pattern **106b** and the first metal pattern **110a** may be formed in each of the first contact holes **104**.

[0080] An uppermost surface of the first barrier metal pattern **106b** may be lower than an upper surface of the first metal pattern **110a**. A first recess **114** may be formed between the first insulating interlayer **102** and the first metal pattern **110a** above the uppermost surface of the first barrier metal pattern **106b**. In the first contact plug **112a**, an upper outer wall of the first metal pattern **110a** may be exposed by the first recess **114**.

[0081] Referring to FIG. **11**, a second insulating interlayer **120** may be formed on the first insulating interlayer **102** and the first contact plug **112a** to fill the first recess **114**. The second insulating interlayer **120** may cover the first insulating interlayer **102** and the first contact plug **112a**. The same insulation material (e.g., the second insulating interlayer) may be formed in the first recess **114** and on the first insulating interlayer **102** and the first contact plug **112a**.

[0082] In example embodiments, voids may not be present in the second insulating interlayer **120** formed in the first recess **114**.

[0083] In some example embodiments, voids may be present in the second insulating interlayer **120** formed in the first recess **114**. In this case, the interconnection structure shown in FIG. **4** may be formed by subsequent processes.

[0084] In some example embodiments, the second insulating interlayer **120** may not fill the first recess **114**. The second insulating interlayer **120** may be formed on the first insulating interlayer **102** and the first contact plug **112a** to cover an inlet of the first recess **114**. Accordingly, the first recess **114** may be empty. In this case, the interconnection structure shown in FIG. **5** may be formed by subsequent processes.

[0085] Referring to FIGS. **12** and **13**, portions of the second insulating interlayer **120** and the first insulating interlayer **102** may be etched to form openings **122**. Processes for forming the openings **122** may include a photo process and an etching process. The etching process may include, e.g., a dry etching process.

[0086] The openings **122** may pass through the second insulating interlayer **120**, and may extend into a portion under the upper surface of the first insulating interlayer **102** in the vertical direction. A first portion of a bottom of each of the openings **122** may expose an upper surface of the first contact plug **112a**. A second portion of the bottom of each of the openings **122** may expose the first insulating interlayer **102**, and may be lower than the upper surface of the first insulating interlayer **102**. The second portion of the bottom of each of the openings **122** may be at an inner portion of the first insulating interlayer **102**.

[0087] In example embodiments, in a plan view, each of the openings 122 may have a line shape extending in one direction or a square shape.

[0088] In example embodiments, the upper surface of the first contact plug 112a exposed by each of the openings 122 may be hardly etched during the etching process. Accordingly, in each of the openings 122, the portion of the bottom exposing the first contact plug 112a may be higher than the other portion of the bottom exposing the first insulating interlayer 102.

[0089] In example embodiments, a lowermost surface of each of the openings 122 may be coplanar with the uppermost surface of the first barrier metal pattern 106b. In some example embodiments, the lowermost surface of each of the openings 122 may be lower than the uppermost surface of the first barrier metal pattern 106b.

[0090] In example embodiments, a lower width of the opening 122 may be less than a width of the first contact plug 112a. Accordingly, in the process for forming the openings 122, an entirety of the upper surface of the first contact plug 112a may not be exposed by each of the openings 122, but a portion of the upper surface of the first contact plug 112a may be exposed by each of the openings 122. Additionally, after forming the openings 122, at least a portion of the second insulating interlayer 120 filling the first recess 114 of the first contact plug 112a may remain without being removed.

[0091] Each of the openings 122 may have a sidewall slope such that a width at the bottom of the opening 122 is less than a width at an upper portion of the opening 122.

[0092] In example embodiments, a portion of the opening 122 positioned lower than the upper surface of the second insulating interlayer 120 may have a sidewall slope such that the width of the opening 122 gradually decreases toward the bottom thereof.

[0093] For example, as shown in FIG. 12, the opening 122 may have a sidewall slope such that the width gradually decreases from an upper portion to the bottom thereof. The sidewall slope may be constant. In this case, the interconnection structure shown in FIG. 1A or FIG. 1B may be formed by subsequent processes. For example, as shown in FIG. 13, a portion of the opening 122 positioned higher than the upper surface of the second insulating interlayer 120 has a vertical sidewall slope, and a portion of the opening 122 positioned lower than the upper surface of the second insulating interlayer 120 may have a sidewall slope such that the width of the opening 122 gradually decreases toward the bottom thereof. In this case, the interconnection structure shown in FIG. 1C may be formed by subsequent processes.

[0094] Referring to FIG. 14, a second barrier metal layer 130 may be conformally formed on sidewalls and bottoms of the openings 122 and on the upper surface of the second insulating interlayer 120. A second metal layer 132 may be formed on the second barrier metal layer 130 to fill the openings 122.

[0095] The second barrier metal layer 130 may include, e.g., titanium, titanium nitride, tantalum, tantalum nitride, etc. The second barrier metal layer 130 may be formed by a chemical vapor deposition process or an atomic layer deposition process.

[0096] In example embodiments, the second barrier metal layer 130 may be formed on the sidewalls and on the bottoms of the openings 122 and the upper surface of the second insulating interlayer 120 to have a uniform thickness.

In some example embodiments, a thickness of the second barrier metal layer 130 formed on the sidewalls of the openings 122 may be less than a thickness on the bottoms of the openings 122 and on the upper surface of the second insulating interlayer 120.

[0097] In example embodiments, the second metal layer 132 may include a material having a resistance lower than a resistance of the first metal pattern 110a. The second metal layer 132 may include, e.g., copper. In some example embodiments, the second metal layer 132 may include, e.g., aluminum, ruthenium, cobalt, nickel, palladium, platinum, silver, or gold.

[0098] Referring to FIG. 15, the second metal layer 124 and the second barrier metal layer 130 may be planarized until the upper surface of the second insulating interlayer 120 is exposed to form a first conductive pattern 134 filling each of the openings 122. The first conductive pattern 134 may include a second barrier metal pattern 130a and a second metal pattern 132a. The planarization process may include, e.g., a chemical mechanical polishing process and/or an etch-back process.

[0099] Since the first conductive pattern 134 is formed in each of the openings 122, the first conductive pattern 134 may have a shape the same as a shape of an inner portion of each of the openings 122.

[0100] Thereafter, a capping layer 140 may be formed on the second metal pattern 132a, the second barrier metal pattern 130a, and the second insulating interlayer 120. The capping layer 140 may cover upper surfaces of the second metal pattern 132a, the second barrier metal pattern 130a, and the second insulating interlayer 120. The capping layer 140 may include an insulation material, e.g., silicon nitride or silicon oxynitride.

[0101] FIG. 16 is a cross-sectional view of an interconnection structure of a semiconductor device according to example embodiments. FIG. 17 is an enlarged cross-sectional view of a portion of FIG. 16.

[0102] The interconnection structure of the semiconductor device shown in FIG. 16 may be the same as the interconnection structure shown in FIG. 1A, except that an adhesive layer is further included.

[0103] Referring to FIGS. 16 and 17, an adhesive layer 116 may be on the first contact plug 112a and the adhesive layer 116 may further be on the first insulating interlayer 102 to fill the first recess 114. The adhesive layer 116 may cover upper surfaces of the first contact plug 112a and the first insulating interlayer 102. A second insulating interlayer 120 may be on the adhesive layer 116.

[0104] The adhesive layer 116 may be disposed between the first insulating interlayer 102 and the second insulating interlayer 120 and the adhesive layer 116 may be further disposed between the first contact plug 112a and the second insulating interlayer 120.

[0105] The adhesive layer 116 may have a thickness less than a thickness of the second insulating interlayer 120, and an upper surface of the adhesive layer 116 may be substantially flat. The adhesive layer 116 may include, e.g., a silicon oxide-based material. The adhesive layer 116 may be provided to improve adhesion between the first and second insulating interlayers 102 and 120.

[0106] In example embodiments, a void may not be present in the adhesive layer 116 formed in the first recess 114. In some example embodiments, a void may be present in the adhesive layer 116 formed in the first recess 114.

[0107] The interconnection structure shown in FIG. 16 may be formed by performing the same processes as processes described with reference to FIGS. 6 to 15, except that a process for forming the adhesive layer 116 may be further performed. First, the processes described with reference to FIGS. 6 to 10 may be performed. Thereafter, before forming the second insulating interlayer 120, the adhesive layer 116 may be formed on the first insulating interlayer 102 and the adhesive layer 116 may be further formed on the first contact plug 112a to fill the first recess 114. Thereafter, the process described with reference to FIGS. 11 to 15 may be performed on the adhesive layer to form the interconnection structure shown in FIG. 16.

[0108] FIG. 18 is a cross-sectional view of a semiconductor device according to example embodiments. FIG. 19 is a cross-sectional view of a semiconductor device according to some example embodiments.

[0109] The semiconductor device shown in FIGS. 18 and 19 may be a DRAM device.

[0110] Referring to FIG. 18, a semiconductor device may be on a substrate 200 including a cell region A and a peripheral circuit region B.

[0111] The substrate 200 may be a wafer substrate including silicon, germanium, silicon-germanium, or a group III-V compound such as GaP, GaAs, GaSb, etc. In some example embodiments, the substrate 200 may be a Silicon On Insulator (SOI) wafer or a Germanium On Insulator (GOI) wafer.

[0112] Isolation layers 202 may be at the substrate 200. The substrate 200 between the isolation layers 202 may be configured as an active region. A first active pattern 200a may be in the cell region A, and a second active pattern 200b may be in the peripheral circuit region B.

[0113] The cell region A of the substrate 200 may include a gate trench extending in a first direction parallel to an upper surface of the substrate 200. A first gate structure may be in the gate trench. In example embodiments, the first gate structure may include a gate insulation layer, a gate electrode, and a capping pattern.

[0114] First impurity regions 205 configured as source/drain regions may be at upper portions of the first active patterns 200a between the first gate structures. The first gate structure and first impurity regions 205 may be configured as a selection transistor. The selection transistor may be a recess channel transistor including the first gate structure buried in the substrate 200.

[0115] An insulation layer pattern 206 may be on the first active pattern 200a, the isolation layer 202, and the first gate structure in the cell region A.

[0116] A bit line structure 220 may be on the insulation layer pattern 206 and the substrate 200. The bit line structure 220 may be on the cell region A of the substrate 200. The bit line structure 220 may contact one of the first impurity regions 205.

[0117] The bit line structure 220 may include a first conductive pattern 220a, a first barrier metal pattern 220b, a first metal pattern 220c, and a hard mask pattern 220d. For example, the first conductive pattern 220a may include polysilicon doped with impurities. The bit line structure 220 may extend in a second direction perpendicular to the first direction, and a plurality of bit line structures may be spaced apart from each other in the first direction. In example embodiments, a first spacer 222 may be on a sidewall of the bit line structure 220.

[0118] Peripheral circuit patterns may be on the peripheral circuit region B of the substrate 200. The peripheral circuit patterns may include, e.g., a peripheral circuit transistor 230, lower wiring, etc. The peripheral circuit transistor 230 may include a second gate structure 224 and second impurity regions 228 configured as source/drain regions. The second impurity regions 228 may be at upper portions of the second active patterns 200b adjacent to both sidewalls of the second gate structure 224. The peripheral circuit transistor 230 may be configured as a sense amplifier, page buffer, command circuit, etc.

[0119] The second gate structure 224 may include a gate insulation layer pattern 207, the first conductive pattern 220a, the first barrier metal pattern 220b, the first metal pattern 220c, and the hard mask pattern 220d. In example embodiments, an upper surface of the second gate structure 224 may be substantially coplanar with an upper surface of the bit line structure 220 in the cell region A. Each of the second gate structures 224 and the bit line structures 220 may include the first conductive pattern 220a, the first barrier metal pattern 220b, the first metal pattern 220c, and the hard mask pattern 220d. Therefore, the second gate structures 224 and the bit line structures 220 may have the same stacked structure.

[0120] In example embodiments, a second spacer 226 may be on a sidewall of the second gate structure 224.

[0121] A first insulating interlayer 232 may occupy a space between the bit line structures 220, and may cover the sidewall of the second gate structure 224 on the peripheral circuit transistor 230. In example embodiments, an upper surface of the first insulating interlayer 232 may be coplanar with upper surfaces of the bit line structure 220 and the second gate structure 224.

[0122] A contact structure may pass through the first insulating interlayer 232 and the insulation layer pattern 206 on the cell region A, and the contact structure may contact a portion of the first impurity region 205. The contact structure may have a structure in which a first lower contact plug 236 and a landing pad 244 are stacked.

[0123] The first lower contact plug 236 may be disposed between the bit line structures 220, and a plurality of first lower contact plugs 236 may be spaced apart from each other. An upper surface of the first lower contact plug 236 may be lower than the upper surface of the bit line structure 220. In example embodiments, the first lower contact plug 236 may include polysilicon.

[0124] The landing pad 244 may be on the first lower contact plug 236. An upper surface of the landing pad 244 may be higher than the upper surface of the bit line structure 220. A first upper insulating pattern 248a may be disposed between the landing pads 244 in the horizontal direction. In example embodiments, an upper portion of the hard mask pattern 220d in the bit line structure 220 may be partially etched through a patterning process, so that an upper portion of the hard mask pattern 220d contacting the first upper insulating pattern 248 may have a recessed shape. In example embodiments, the landing pad 244 may include tungsten. In example embodiments, a barrier metal pattern may be further included at an interface between the landing pad 244 and the first lower contact plug 236.

[0125] A second lower contact plug 242 may pass through the first insulating interlayer 232 on the peripheral circuit region B, and the second lower contact plug 242 may be

connected to the second impurity regions **228** or the first metal pattern **220c** of the second gate structure **224**.

[0126] A second conductive pattern **246** may be on the first insulating interlayer **232** on the peripheral circuit region B. The second conductive patterns **246** may be wiring lines for electrically connecting the peripheral circuits. The second conductive patterns **246** may include the same material as the landing pad **244** on the cell region A.

[0127] A second upper insulating pattern **248b** may be between the second conductive patterns **246**. Upper surfaces of the second conductive patterns **246** and the second upper insulating pattern **248b** may be coplanar with upper surfaces of the landing pad **244** and the first upper insulating pattern **248a**.

[0128] A first etch stop layer **250** may be on the upper surfaces of the landing pad **244**, the second conductive patterns **246**, and the first and second upper insulating patterns **248a** and **248b**. The first etch stop layer **250** may include, e.g., silicon nitride.

[0129] A cell capacitor may pass through the first etch stop layer **250**, and may contact the landing pad **244**. The cell capacitor may include a plurality of lower electrodes **262**, a dielectric layer pattern **270a**, and an upper electrode **272a**. In addition, the cell capacitor may further include a lower support layer pattern **254a** and an upper support layer pattern **258a** for supporting the lower electrodes **262**.

[0130] The lower electrodes **262** may pass through the first etch stop layer **250**, and the lower electrodes **262** may be on the upper surface of each of the landing pads **244**, respectively. In example embodiments, the lower electrodes **262** may have a pillar shape. In some example embodiments, the lower electrodes **262** may have a cylindrical shape.

[0131] In example embodiments, the lower electrodes **262** may include a metal, e.g., Ti, W, Ni, or Co, or a metal nitride, e.g., TiN, TiSiN, TiAlN, TaN, TaSiN, or WN. For example, the lower electrodes **262** may include TiN.

[0132] Each of the lower support layer pattern **254a** and the upper support layer pattern **258a** may be disposed between the lower electrodes **262**. Each of the lower support layer pattern **254a** and the upper support layer pattern **258a** may be connected to sidewalls of the lower electrodes **262**, so that each of the lower support layer pattern **254a** and the upper support layer pattern **258a** may support the sidewalls of the lower electrodes **262**. The lower support layer pattern **254a** and the upper support layer pattern **258a** may include an insulation material, e.g., silicon nitride or silicon oxynitride.

[0133] In some example embodiments, at least one of the upper support layer pattern **258a** and the lower support layer pattern **254a** may not be included. In some example embodiments, two or more first lower support layer patterns may be below the upper support layer pattern **258a**.

[0134] The dielectric layer pattern **270a** may conform to surfaces of the lower electrode **262**, the lower support layer pattern **254a**, the upper support layer pattern **258a**, and the first etch stop layer **250**. The dielectric layer pattern **270a** may include a high-k dielectric layer. In example embodiments, the high-k dielectric layer may include a metal layer, e.g., a hafnium oxide layer (HfO₂), a zirconium oxide layer (ZrO₂), an aluminum oxide layer (Al₂O₃), or a lanthanum oxide layer (La₂O₅).

[0135] The upper electrode **272a** may be on the dielectric layer pattern **270a**. In example embodiments, the upper electrode **272a** may include metal nitride. The metal nitride

may include, e.g., titanium nitride (TiN), titanium silicon nitride (TiSiN), titanium aluminum nitride (TiAlN), tantalum nitride (TaN), tantalum silicon nitride (TaSiN), tantalum aluminum nitride (TaAlN), tungsten nitride (WN), or the like.

[0136] In example embodiments, the upper electrode **272a** may conform to the dielectric layer pattern **270a**, and may not completely occupy a space between the lower electrodes **262**.

[0137] The dielectric layer pattern **270a** and the upper electrode **272a** may be only on the cell region A of the substrate **200**. The dielectric layer pattern **270a** and the upper electrode **272a** may not be on the peripheral circuit region B of the substrate **200**. Accordingly, the dielectric layer pattern **270a** and the upper electrode **272a** B may be stacked on the upper surface of the first etch stop layer **250** at an edge portion of the cell region A adjacent to the peripheral circuit region B.

[0138] A first plate pattern **280** may be on the surface of the upper electrode **272a**. In example embodiments, the first plate pattern **280** may be only on the cell region A. The first plate pattern **280** may include, e.g., silicon germanium.

[0139] In example embodiments, the first plate pattern **280** may occupy a space between the upper electrodes **272a**, and the first plate pattern **280** may cover an uppermost surface of the upper electrode **272a**. Additionally, the first plate pattern **280** may be on the edge of the cell region A and on a sidewall of a last cell capacitor on the edge of the cell region A adjacent to the peripheral circuit region B. The first plate pattern **280** may not be on the peripheral circuit region B.

[0140] In example embodiments, the first plate pattern **280** may include a first portion, a second portion, and a third portion. The first portion may be positioned on the upper electrode **272a** on the cell region A, and an upper surface of the first portion may be substantially flat. The second portion may cover the last cell capacitor on the cell region A adjacent to the peripheral circuit region B, and may have a vertical surface perpendicular to the surface of the substrate **200**. The third portion may be connected to the second portion, and an upper surface of the third portion may be substantially flat. The third portion may be positioned on the edge of the cell region A adjacent to the peripheral circuit region.

[0141] An upper metal pattern **282** and a second etch stop pattern **284** may be stacked on the first plate pattern **280**.

[0142] A second insulating interlayer **286** may be on the second etch stop pattern **284** and the first etch stop layer **250**. An upper surface of the second insulating interlayer **286** may be substantially flat. The upper surface of the second insulating interlayer **286** may be higher than an upper surface of the second etch stop pattern **284**.

[0143] A first contact plug **302** may pass through the second insulating interlayer **286** and the first etch stop layer **250** on the peripheral region, and may contact each of the second conductive patterns **246**. The first contact plug **302** may be disposed in the first contact hole **292**.

[0144] A second contact plug **304** may pass through the second insulating interlayer **286** and the second etch stop pattern **284** on the cell region, and may contact the upper metal pattern **282**. The second contact plug **304** may be disposed in the second contact hole **294** passing through the second insulating interlayer **286** and the second etch stop pattern **284** and exposing the upper metal pattern **282**.

[0145] The second contact plug 304 may be disposed over the cell capacitor. Accordingly, a bottom of the second contact plug 304 may be higher than a bottom of the first contact plug 302. The upper surface of the second contact plug 304 and the upper surface of the first contact plug 302 may be coplanar with each other. A height in the vertical direction of the first contact plug 302 may be greater than a height in the vertical direction of the second contact plug 304.

[0146] Each of the first and second contact plugs 302 and 304 may include a second barrier metal pattern 296a and a second metal pattern 298. The second barrier metal pattern 296a may cover a portion of a sidewall of the second metal pattern 298 and a bottom of the second metal pattern 298.

[0147] An upper surface of the second metal pattern 298 may be substantially coplanar with the upper surface of the second insulating interlayer 286. Additionally, an uppermost surface of the second barrier metal pattern 296a may be lower than the upper surface of the second metal pattern 298. The second barrier metal pattern 296a may not be on an upper sidewall of the second metal pattern 298.

[0148] A first recess 300 may be disposed between the upper sidewall of the second metal pattern 298 and the second insulating interlayer 286, and the first recess 300 may have a shape surrounding the upper sidewall of the second metal pattern 298. The first recess 300 may be disposed above the uppermost surface of the second barrier metal pattern 296a.

[0149] A third insulating interlayer 306 may be on the first contact plug 302, the second contact plug 304, and the second insulating interlayer 286.

[0150] In example embodiments, the third insulating interlayer 306 may occupy the first recess 300. A silicon oxide-based material may occupy the first recess 300. The first recess 300 may not be occupied with nitride material, such as silicon nitride.

[0151] In example embodiments, the third insulating interlayer 306 in the first recess 300 may not include voids. In some example embodiments, the third insulating interlayer 306 in the first recess 300 may include voids. In some example embodiments, the third insulating interlayer 306 may not occupy the first recess 300.

[0152] First and second openings 310 and 312 may pass through the third insulating interlayer 306, and may extend into a portion under the upper surface of the second insulating interlayer 120 in the vertical direction. The first openings 310 may be disposed on the peripheral circuit region B, and the first contact plug 302 may be exposed by at least a portion of the first openings 310. The second opening 312 are disposed on the cell region A, and the second contact plug 304 may be exposed by at least a portion of the second opening 312.

[0153] Each of the first and second openings 310 and 312 may have a sidewall slope such that a lower width of each of the first and second openings 310 and 312 may be less than an upper width of each of the first and second openings 310 and 312. A first conductive pattern structure 320 may be in each of the first openings 310, and a second conductive pattern structure 324 may be in the second opening 312.

[0154] A first bottom surface of the first conductive pattern structure 320 may contact an upper surface of the first contact plug 302. A second bottom surface of the first conductive pattern structure 320 may contact the second insulating interlayer 286. The third bottom surface of the

second conductive pattern structure 324 may contact an upper surface of the second contact plug 304. The fourth bottom surface of the second conductive pattern structure 324 may contact the second insulating interlayer 286.

[0155] The first bottom surface of the first conductive pattern structure 320 may be positioned higher than the second bottom surface of the first conductive pattern structure 320. The third bottom surface of the second conductive pattern structure 324 may be positioned higher than the fourth bottom surface of the second conductive pattern structure 324. The second and fourth bottom surfaces may be in the second insulating interlayer 286. In example embodiments, the second and fourth bottom surfaces may be coplanar with or lower than a bottom of the first recess 300.

[0156] Each of the first and second conductive pattern structures 320 and 324 may include a third barrier metal pattern 314 and a third metal pattern 316. The third barrier metal pattern 314 may cover a sidewall and a bottom of the third metal pattern 316.

[0157] The third metal pattern 316 may be on the third barrier metal pattern 314 to occupy the first and second openings 310 and 312. Upper surfaces of the third metal pattern 316 and the third barrier metal pattern 314 may be substantially coplanar with an upper surface of the third insulating interlayer 306.

[0158] In example embodiments, a width of an uppermost portion of each of the first and second conductive pattern structures 320 and 324 may be less than a width of a lowermost portion of the first and second conductive pattern structures 320 and 324. In example embodiments, at least a lower portion of each of the first and second conductive pattern structures 320 and 324 may have a sidewall slope such that the width of the lower portion of each of the first and second conductive pattern structures 320 and 324 gradually decreases downward.

[0159] A portion of the first conductive pattern structure 320 lower than the upper surface of the second insulating interlayer 286 is referred to in the following disclosure as a fourth portion. In the claims, the recitation of a “fourth portion” without more is not intended to incorporate this definition therein. The fourth portion of the first conductive pattern structure 320 may be adjacent to the first contact plug 302 to be spaced apart from the first contact plug 302 in the horizontal direction.

[0160] In a cross-sectional view, the upper sidewall of the first contact plug 302 and the fourth portion of the first conductive pattern structure 320 may face toward each other in the horizontal direction.

[0161] A distance in the horizontal direction between an uppermost sidewall of the second metal pattern 298 of the first contact plug 302 and the first conductive pattern structure 320 may be greater than a distance in the horizontal direction between an uppermost sidewall of the second barrier metal pattern 296a of the first contact plug 302 and the first conductive pattern structure 320.

[0162] In example embodiments, an upper portion of the fourth portion of the first conductive pattern structure 320 may face the sidewall of the second metal pattern 298, and a lower portion of the fourth portion of the first conductive pattern structure 320 may face the sidewall of the second barrier metal pattern 296a. In some example embodiments, the fourth portion of the second conductive pattern structure 324 may face only the sidewall of the second metal pattern 298.

[0163] The distance in the horizontal direction between the uppermost sidewall of the second barrier metal pattern 296a of the first contact plug 302 and the first conductive pattern structure 320 is referred to in the following disclosure as a first distance. In the claims, the recitation of a “first distance” without more is not intended to incorporate this definition therein. Since the uppermost portion of the second barrier metal pattern 296a has a recessed shape, the first distance may be greater than a first distance when an uppermost portion of a second barrier metal pattern is not recessed.

[0164] A region having the first distance may be a weak region where insulation breakdown of insulation material between the first contact plug 302 and the first conductive pattern structure 320 easily occurs, and thus reliability of the semiconductor device may be compromised. However, since an interface between the insulation materials is not positioned at the region having the first distance, reliability may be improved.

[0165] The second conductive pattern structure 324 may extend in one direction (e.g., longitudinal direction). In example embodiments, as shown in FIG. 18, two or more second contact plugs 304 arranged in a width direction perpendicular to the longitudinal direction may contact the bottom of the same second conductive pattern structure 324. In some example embodiments, as shown in FIG. 19, only one second contact plug 304 in the width direction may contact the bottom of the second conductive pattern structure 324.

[0166] A capping layer 330 may be on the first conductive pattern structure 320, the second conductive pattern structure 324, and the third insulating interlayer 306. The capping layer 330 may cover the first conductive pattern structure 320, the second conductive pattern structure 324, and the third insulating interlayer 306. The capping layer 330 may include an insulation material, e.g., silicon nitride or silicon oxynitride.

[0167] As described above, a portion of the first conductive pattern structure formed on the peripheral circuit region in the DRAM device may face the upper portion of the first contact plug disposed below the first conductive pattern structure in the horizontal direction. Since the upper portion of the barrier metal pattern included in the first conductive pattern structure has the recessed shape, a minimum distance between the first conductive pattern structure and the first contact plug may be increased. Additionally, the interface of the insulation material may not be positioned at a region having the minimum distance. Therefore, the DRAM device may have high reliability.

[0168] FIGS. 20 to 34 are cross-sectional views illustrating a method of manufacturing a semiconductor device according to example embodiments.

[0169] Referring to FIG. 20, an isolation layer 202 for defining an active region may be formed on the substrate 200 including a cell region A and a peripheral circuit region B. Accordingly, a first active pattern 200a may be formed on the cell region A, and the second active pattern 200b may be formed on the peripheral circuit region B.

[0170] Select transistors may be formed on the cell region A of the substrate 200. Each of the selection transistors may include a first gate structure and first impurity regions 205. In example embodiments, a portion of the substrate 200 may be etched to form a gate trench, and the first gate structure may be formed in the gate trench. For example, the first gate

structure may include a gate insulation layer, a gate electrode, and a capping pattern sequentially stacked.

[0171] An insulation layer pattern 206 may be formed on the cell region A of the substrate 200. A recess may be formed on the first active pattern 200a between the insulation layer patterns 206. An upper surface of the first impurity region 205 may be exposed by a bottom of the recess. Additionally, a gate insulation layer may be formed on the peripheral circuit region B of the substrate 200.

[0172] Bit line structures 220 may be formed on the insulation layer pattern 206 and the recess. The bit line structure 220 may include a first conductive pattern 220a, a first barrier metal pattern 220b, a first metal pattern 220c, and a hard mask pattern 220d. In example embodiments, a first spacer 222 may be formed on a sidewall of the bit line structure 220.

[0173] When the bit line structure 220 is formed on the cell region A, a second gate structure 224 may be formed on the peripheral circuit region B of the substrate 200 together. The second gate structure 224 may include a gate insulation layer pattern 207, a first conductive pattern 220a, a first barrier metal pattern 220b, a first metal pattern 220c, and a hard mask pattern 220d sequentially stacked. In example embodiments, a second spacer 226 may be formed on a sidewall of the second gate structure 224. After forming the second gate structure 224, second impurity regions 228 configured as source/drain regions may be formed at an upper portion of the second active pattern 200b adjacent to both sidewalls of the second gate structure 224.

[0174] Accordingly, a peripheral circuit transistor 230 including a second gate structure 224 and second impurity regions 228 may be formed on the peripheral circuit region B of the substrate 200. The peripheral circuit transistor 230 may form a peripheral circuit.

[0175] A first insulating interlayer 232 may be formed on the substrate 200 to cover the bit line structures 220 and the peripheral circuit transistor 230. Thereafter, an upper surface of the first insulating interlayer 232 may be planarized. In example embodiments, the first insulating interlayer 232 may be planarized until upper surfaces of the bit line structure 220 and the second gate structure 224 are exposed.

[0176] A portion of the first insulating interlayer 232 between the bit line structures 220 may be etched to form a first lower contact hole 234 exposing a portion of the first impurity region 205 of the substrate 200.

[0177] Referring to FIG. 21, a first conductive layer may be formed on the first insulating interlayer 232 to fill the first lower contact hole 234. Thereafter, the first conductive layer may be partially etched by an etch-back process to form a first lower contact plug 236 filling a lower portion of the first lower contact hole 234. In the etch-back process, all of the first conductive layer formed on the first insulating interlayer 232 may be removed. In example embodiments, the first lower contact plug 236 may include polysilicon.

[0178] Thereafter, a portion of the first insulating interlayer 232 on the peripheral circuit region B may be etched to form a second lower contact hole 238 exposing the second impurity region 228 in the second active pattern 200b.

[0179] Referring to FIG. 22, a second conductive layer 240 may be formed to cover the bit line structure 220, the first lower contact plug 236, and the first insulating interlayer 232. The second conductive layer may fill the first lower contact hole 234 and the second lower contact hole 238. In example embodiments, the second conductive layer

240 may include tungsten. In some example embodiments, before forming the second conductive layer 240, a process for forming a barrier metal layer may be further performed.

[0180] In example embodiments, before forming the second conductive layer 240, a process for forming a metal silicide pattern on the first lower contact plug 236 may be further performed. In this case, the metal silicide pattern may also be formed on the surface of the substrate 200 exposed by a bottom of the second lower contact hole 238 on the peripheral circuit region B. By the above process, a second lower contact plug 242 may be formed in the second lower contact hole 238 on the peripheral circuit region B.

[0181] Referring to FIG. 23, the second conductive layer 240 on the cell region A may be patterned to form a landing pad 244 contacting an upper surface of the first lower contact plug 236. The second conductive layer 240 on the peripheral circuit region B may be patterned to form second conductive patterns 246.

[0182] Patterning process for forming the landing pad 244 on the cell region A and the second conductive patterns 246 on the peripheral circuit region B may be performed by different photolithography processes or the same single photo lithography process.

[0183] In the patterning process, an upper portion of the hard mask pattern 220d included in the bit line structure 220 may be partially etched, so that the upper portion of the bit line structure 220 may have a recessed shape.

[0184] Referring to FIG. 24, an upper insulation layer may be formed on the landing pad 244, the second conductive patterns 246, and the first insulating interlayer 232. The upper insulation layer may be planarized until upper surfaces of the landing pad 244 and the second conductive patterns 246 are exposed. The planarization process may include, e.g., a chemical mechanical polishing process.

[0185] Accordingly, a first upper insulating pattern 248a may be formed in a gap between the landing pads 244, and a second upper insulating pattern 248b may be formed in a gap between the second conductive patterns 246. Upper surfaces of the landing pad 244, the second conductive pattern 246, and the first and second upper insulating patterns 248a and 248b may be substantially coplanar with each other. Additionally, upper surfaces of the landing pad 244, the second conductive pattern 246, and the first and second upper insulating patterns 248a and 248b may be substantially flat.

[0186] A first etch stop layer 250 may be formed on the landing pad 244, the second conductive pattern 246, and the first and second upper insulating patterns 248a and 248b. The first etch stop layer 250 may include, e.g., silicon nitride, silicon oxynitride, etc.

[0187] A first mold layer 252, a lower support layer 254, a second mold layer 256, and an upper support layer 258 may be sequentially deposited on the first etch stop layer 250.

[0188] The first and second mold layers 252 and 256 may include a material having an etch selectivity with respect to the lower support layer 254 and the upper support layer 258. The first mold layer 252 and the second mold layer 256 may include, e.g., silicon oxide, and the lower support layer 254 and the upper support layer 258 may include, e.g., silicon nitride.

[0189] In this example embodiment, two support layers may be included in the semiconductor device, but may not be limited thereto. In some example embodiments, a single

support layer or three or more support layers may be included in the semiconductor device, depending on a structure of a cell capacitor. In some example embodiments, the support layer may not be formed, and thus one mold layer may be included in the semiconductor.

[0190] An etch mask may be formed on the upper support layer 258. The upper support layer 258, the second mold layer 256, the lower support layer 254, and the first mold layer 252 and the first etch stop layer 250 may be etched using the etch mask to form first holes. The etching process may include an anisotropic etching process. By the above process, first holes 260 may expose the upper surface of the landing pad 244 on the cell region A of the substrate 200.

[0191] Thereafter, a lower electrode layer may be formed to fill the first holes 260. The lower electrode layer may be planarized until an upper surface of the upper support layer 258 is exposed to form lower electrodes 262 filling the first holes 260. In this case, each of the lower electrodes 262 may have a pillar shape. The lower electrode 262 may include a metal, e.g., Ti, W, Ni, or Co, or a metal nitride such as TiN, TiSiN, TiAlN, TaN, TaSiN, or WN.

[0192] Referring to FIG. 25, a first mask pattern may be formed on the upper support layer 258 and the lower electrode 262. The first mask pattern may include amorphous carbon or polysilicon. The first mask pattern may include holes.

[0193] The upper support layer 258 may be anisotropically etched using the first mask pattern to form second holes exposing the upper portion of the second mold layer 256. Accordingly, an upper support layer pattern 258a may be formed on the cell region A of the substrate 200. Thereafter, the second mold layer 256 may be wet etched by supplying etchant through the second holes. Accordingly, the lower support layer 254 may be exposed. Additionally, upper sidewalls of the lower electrodes 262 may be exposed.

[0194] A portion of the lower support layer 254 may be etched to form third holes exposing an upper portion of the first mold layer 252. Accordingly, the lower support layer pattern 254a may be formed. The first mold layer 252 may be wet etched by supplying etchant through the third holes.

[0195] The upper support layer pattern 258a and the lower support layer pattern 254a may be connected to each other while contacting a portion of an outer wall of the lower electrode 262. Accordingly, sidewalls of the lower electrode 262 may be supported by the upper support layer pattern 258a and the lower support layer pattern 254a, so that a leaning of the lower electrode 262 may be prevented. Additionally, the surface of the lower electrode 262 may be exposed.

[0196] Thereafter, a dielectric layer 270 may be conformally formed along the surfaces of the lower electrode 262, the lower and upper support layer patterns 254a and 258a, and the first etch stop layer 250. An upper electrode layer 272 including metal may be formed on the dielectric layer 270.

[0197] The dielectric layer 270 may include a high dielectric layer. In example embodiments, the high-k dielectric layer may include a hafnium oxide layer (HfO₂), a zirconium oxide layer (ZrO₂), an aluminum oxide layer (Al₂O₃), or a lanthanum oxide layer (La₂O₅).

[0198] In example embodiments, the upper electrode layer 272 may include a metal nitride.

[0199] Referring to FIG. 26, a first plate layer may be formed on the upper electrode layer 272. The first plate layer

may include, e.g., a silicon germanium layer. The silicon germanium layer may be doped with, e.g., P-type impurities or N-type impurities.

[0200] In example embodiments, the silicon germanium layer may be formed by an atomic layer deposition process or a chemical vapor deposition process using a silicon source gas, a germanium source gas, and a dopant gas as deposition gases.

[0201] The silicon source gas may include, e.g., silane (SiH₄) or dichloro silane (SiH₂Cl₂). The germanium source gas may include, e.g., germanium (GeH₄) or germanium tetrachloride (GeCl₄). The dopant gas may include, e.g., borane (BH₃), boron chloride (BCl₃), phosphine (PH₃), phosphorus chloride (PCl₃), etc. Thereafter, the first plate layer may be crystallized by a heat treatment process.

[0202] Since the upper electrode layer 272 includes metal, it is not easy to increase a thickness of the upper electrode layer 272. Therefore, if the upper electrode is formed only using the upper electrode layer 272, a second contact plug subsequently formed may unintentionally pass through the upper electrode and extend into the lower electrode 262. Therefore, the first plate layer may serve as a buffer layer for decreasing the likelihood of such a defect in subsequent processes.

[0203] The first plate layer may be conformally formed on the upper electrode layer 272 on the cell region A and the peripheral circuit region B to fill a gap between the upper electrode layers 272.

[0204] An upper metal layer and a second etch stop layer may be formed on the first plate layer. The upper metal layer may include, e.g., tungsten. The second etch stop layer may include, e.g., silicon nitride or silicon oxynitride.

[0205] Thereafter, the second etch stop layer, the upper metal layer, the first plate layer, the upper electrode layer, and the dielectric layer on the peripheral circuit region B may be etched to form a dielectric layer pattern 270a, an upper electrode 272a, a first plate pattern 280, an upper metal pattern 282, and a second etch stop pattern 284. A stacked structure of the dielectric layer pattern 270a, the upper electrode 272a, the first plate pattern 280, the upper metal pattern 282, and the second etch stop pattern 284 may be formed only on the cell region A.

[0206] The first plate pattern 280 may have a first portion having an upper surface higher than the upper surface of the lower electrode 262 on the cell region A, a second portion covering a last cell capacitor in the cell region A adjacent to the peripheral circuit region B, and a third portion connected to the second portion and disposed at the edge of the cell region. The second portion of the first plate pattern 280 may have a vertical surface perpendicular to the surface of the substrate 200.

[0207] Referring to FIG. 27, a second insulating interlayer 286 may be deposited on the first plate pattern 280 and the first etch stop layer 250. The second insulating interlayer 286 may include a silicon oxide-based material. The second insulating interlayer 286 may include, e.g., a TEOS layer.

[0208] An upper surface of the second insulating interlayer 286 may be higher than an upper surface of the second etch stop pattern 284. The upper surface of the second insulating interlayer 286 may be substantially flat. In example embodiments, a planarization process of the upper surface of the second insulating interlayer 286 may be

further performed. The planarization process may include, e.g., a chemical mechanical polishing process and/or an etch-back process.

[0209] Referring to FIG. 28, the second insulating interlayer 286 and the first etch stop layer 250 on the peripheral circuit region B may be etched to form first contact holes exposing the second conductive pattern 246. The second insulating interlayer 286 and the second etch stop pattern 284 on the cell region A may be etched to form second contact holes 294 exposing the upper metal pattern 282.

[0210] In some example embodiments, example embodiments, the second contact hole 294 may have a sidewall slope such that a width of the second contact hole 294 may decrease from an upper portion of the second contact hole 294 to a bottom of the second contact hole 294. The sidewall slope may be constant. Accordingly, the second contact hole 294 may have a lower width less than an upper width.

[0211] A second barrier metal layer may be formed on the surfaces of the first contact holes 292 and the second contact holes 294 and the second insulating interlayer 286. A second metal layer may be formed on the second barrier metal layer to fill the first contact holes 292 and the second contact holes 294.

[0212] The second barrier metal layer may include, e.g., titanium, titanium nitride, tantalum, tantalum nitride, etc. The second metal layer may include, e.g., tungsten. In some example embodiments, the second metal layer may include, e.g., aluminum, copper, ruthenium, cobalt, nickel, palladium, platinum, silver, or gold.

[0213] The second metal layer and the second barrier metal layer may be planarized until the upper surface of the second insulating interlayer 286 is exposed to form a preliminary barrier metal pattern 296 and a second metal pattern 298a. The planarization process may include, e.g., a chemical mechanical polishing process. The preliminary barrier metal pattern 296 and the second metal pattern 298a may be formed in each of the first contact holes 292 and the second contact holes 294.

[0214] Referring to FIG. 29, an upper portion of the preliminary barrier metal pattern 296 may be partially etched to form a second barrier metal pattern 296a in each of the first contact hole 292 and the second contact hole 294.

[0215] Accordingly, a first contact plug 302 including the second barrier metal pattern 296a and the second metal pattern 298 may be formed in the first contact hole 292, and a second contact plug 304 including a second barrier metal pattern 296a and a second metal pattern 298 may be formed in the second contact hole 294.

[0216] In the first contact plug 302, an uppermost surface of the second barrier metal pattern 296a may be lower than an upper surface of the second metal pattern 298. Additionally, in the second contact plug 304, the uppermost surface of the second barrier metal pattern 296a may be lower than the upper surface of the second metal pattern 298. A first recess 300 may be formed between the second insulating interlayer 286 and the second metal pattern 298 above the upper surface of the second barrier metal pattern 296a.

[0217] Referring to FIG. 30, a third insulating interlayer 306 may be formed on the second insulating interlayer 286 and the first and second contact plugs 302 and 304 to fill the first recess 300. The third insulating interlayer 306 may cover the second insulating interlayer 286 and the first and second contact plugs 302 and 304. The same insulation material (e.g., the third insulating interlayer) may be formed

in the first recess **300** and on the second insulating interlayer **286**, and the first and second contact plugs **302** and **304**.

[0218] Referring to FIGS. **31** and **32**, upper portions of the third insulating interlayer **306** and the second insulating interlayer **286** on the peripheral circuit region **B** may be etched to form a first opening **310** exposing at least a portion of the first contact plug **302**. A portion of the third insulating interlayer **306** and the second insulating interlayer **286** on the cell region **A** may be etched to form a second opening **312** exposing at least a portion of the second contact plug **304**.

[0219] The process for forming the first and second openings **310** and **312** may include a photo process and an etching process. The etching process may include, e.g., a dry etching process.

[0220] The first and second openings **310** and **312** may pass through the third insulating interlayer **306**, and extend into a portion under the upper surface of the second insulating interlayer **286** in the vertical direction.

[0221] In example embodiments, in a plan view, each of the first and second openings **310** and **312** may have a line shape or a square shape extending in one direction.

[0222] In example embodiments, as shown in FIG. **31**, a plurality of second contact plugs **304** arranged in the width direction may be exposed by a bottom of the second opening **312**. In some example embodiments, as shown in FIG. **32**, one second contact plug **304** in the width direction may be exposed by a bottom of the second opening **312**. Each of the first and second openings **310** and **312** may have a sidewall slope such that a lower width of each of the first and second openings **310** and **312** is less than an upper width of each of the first and second openings **310** and **312**.

[0223] In example embodiments, in each of the first and second openings **310** and **312**, at least a portion lower than the upper surface of the second insulating interlayer **286** may have a sidewall slope such that a width is gradually decreased toward the bottom.

[0224] In the etching process for forming the first and second openings **310** and **312**, the first and second contact plugs **302** and **304** may be hardly etched. Accordingly, the bottom of the first opening **310** exposing each of the first contact plugs **302** may be higher than the bottom of the first opening **310** exposing the second insulating interlayer **286**, and the bottom of the second opening **312** exposing the second contact plugs **304** may be higher than the bottom of the second opening **312** exposing the second insulating interlayer **286**.

[0225] In example embodiments, at least one first opening **310** may expose only the second insulating interlayer **286** may be disposed between the two first openings **310** exposing first contact plugs **302**.

[0226] In example embodiments, at least one first opening **310** may be disposed between neighboring first contact plugs **302** in the horizontal direction. Only the second insulating interlayer **286** may be exposed by a lower sidewall of the first opening **310** disposed between the neighboring first contact plugs **302** in the horizontal direction.

[0227] Referring to FIGS. **33** and **34**, a third barrier metal layer may be conformally formed on sidewalls and on bottom surfaces of the first and second openings **310** and **312** and on the upper surface of the third insulating interlayer **306**. A third metal layer may be formed on the third barrier metal layer to fill the first and second openings **310** and **312**.

[0228] The third barrier metal layer may include, e.g., titanium, titanium nitride, tantalum, tantalum nitride, etc. In example embodiments, the third metal layer may include a material having a resistance lower than a resistance of the second metal pattern **298**. For example, the third metal layer may include copper. In some example embodiments, the third metal layer may include aluminum, ruthenium, cobalt, nickel, palladium, platinum, silver, or gold.

[0229] Thereafter, the third metal layer and the third barrier metal layer may be planarized until an upper surface of the third insulating interlayer **306** is exposed to form a third barrier metal pattern **314** and a third metal pattern **316**. The third barrier metal pattern **314** and the third metal pattern **316** may be formed in each of the first opening **210** and the second opening. The planarization process may include, e.g., a chemical mechanical polishing process and/or an etch-back process.

[0230] In example embodiments, as shown in FIG. **33**, the second conductive pattern structure **324** may contact at least two second contact plugs **304** in the width direction. In this case, the semiconductor device shown in FIG. **18** may be manufactured by subsequent processes. In some example embodiments, as shown in FIG. **34**, the second conductive pattern structure **324** may contact only one second contact plug **304** arranged in the width direction. In this case, the semiconductor device shown in FIG. **19** may be manufactured by subsequent processes.

[0231] The first conductive pattern structure **320** may include a first bottom surface contacting a first contact plug **302** and a second bottom surface contacting the second insulating interlayer **286**. The second bottom surface of the first conductive pattern structure **320** may be lower than the first bottom surface of the first conductive pattern structure **320**.

[0232] A portion of the first conductive pattern structure **320** positioned lower than the upper surface of the second insulating interlayer **286** is referred to in the following disclosure as a fourth portion, and the fourth portion of the first conductive pattern structure **320** and an upper sidewall of the first contact plug **302** may face toward each other in the horizontal direction. In the claims, the recitation of a “fourth portion” without more is not intended to incorporate this definition therein.

[0233] In example embodiments, the sidewall of the second metal pattern **298** may be positioned higher than the uppermost surface of the second barrier metal pattern **296a** in the first contact plug **302**, and the sidewall of the second metal pattern **298** and the fourth portion of the conductive pattern structure **320** may face toward each other in the horizontal direction.

[0234] Referring to FIGS. **19** and **20** again, a capping layer **330** may be formed on the first and second conductive pattern structures **320** and **324** and the third insulating interlayer **306**. The capping layer **330** may cover upper surfaces of the first and second conductive patterns **320** and **324** and the third insulating interlayer **306**. The capping layer **330** may include an insulation material, e.g., silicon nitride or silicon oxynitride.

[0235] FIG. **35** is a cross-sectional view of a semiconductor device according to example embodiments.

[0236] The semiconductor device shown in FIG. **35** may be substantially the same as the semiconductor device shown in FIG. **19**, except that an adhesive layer is further included.

[0237] Referring to FIG. 35, an adhesive layer 340 may occupy the first recess 300, and the adhesive layer 340 may further cover upper surfaces of the first and second contact plugs 302 and 304 and the second insulating interlayer 286. A third insulating interlayer 306 may be on the adhesive layer 340.

[0238] In example embodiments, voids may not be present in the adhesive layer 340 in the first recess 300. In some example embodiments, voids may be present in the adhesive layer 340 in the first recess 300.

[0239] Processes for manufacturing the semiconductor device shown in FIG. 35 may be substantially the same as processes as those described with reference to FIGS. 20 to 34, except for a process for forming the adhesive layer. First, the process described with reference to FIGS. 20 to 29 may be performed. Thereafter, before forming the third insulating interlayer 306, the adhesive layer 340 may be formed on the second insulating interlayer 286 and the first and second contact plugs 302 and 304 to fill the first recess 300. Thereafter, the processes described with reference to FIGS. 30 to 34 may be performed on the adhesive layer 340 to manufacture the semiconductor device.

[0240] The interconnection structure included in the semiconductor device may have high reliability, so the semiconductor device may have high reliability.

[0241] While the present inventive concepts have been shown and described with reference to example embodiments thereof, it will be understood by those of ordinary skill in the art that various changes in form and details may be made thereto without departing from the scope of the present inventive concepts as set forth by the following claims.

What is claimed is:

1. A semiconductor device, comprising:
 - a substrate;
 - a first insulating interlayer on the substrate;
 - a plurality of first contact plugs passing through the first insulating interlayer, each of the plurality of first contact plugs including:
 - a first metal pattern; and
 - a first barrier metal pattern covering a portion of a sidewall of the first metal pattern and further covering a bottom of the first metal pattern, and the first barrier metal pattern further exposing an upper sidewall of the first metal pattern;
 - a second insulating interlayer that is on the first insulating interlayer and is further on the plurality of first contact plugs; and
 - a plurality of first conductive patterns passing through the second insulating interlayer and extending below an upper surface of the first insulating interlayer in a vertical direction, each of the plurality of first conductive patterns having:
 - a first bottom surface contacting a portion of an upper surface of a respective first contact plug of the plurality of first contact plugs; and
 - a second bottom surface contacting the first insulating interlayer and being lower than the first bottom surface.
2. The semiconductor device of claim 1, wherein an upper surface of the first insulating interlayer and an upper surface of the first metal pattern are coplanar with each other, and a first recess is between the first insulating interlayer and the first metal pattern.

3. The semiconductor device of claim 2, wherein the second insulating interlayer occupies the first recess.

4. The semiconductor device of claim 2, further comprising an adhesive layer that is between the first insulating interlayer and the second insulating interlayer and is further between the plurality of first contact plugs and the second insulating interlayer, and

wherein the adhesive layer occupies the first recess.

5. The semiconductor device of claim 1, wherein an uppermost surface of the first barrier metal pattern is lower than an upper surface of the first metal pattern, and

wherein each of the plurality of first conductive patterns contacts each of:

- the upper surface of the first metal pattern;
- the upper sidewall of the first metal pattern; and
- an upper surface of the first barrier metal pattern.

6. The semiconductor device of claim 1, wherein the second bottom surface of each first conductive pattern is coplanar with or lower than an upper surface of the first barrier metal pattern of the respective first contact plug.

7. The semiconductor device of claim 1, wherein at least a portion of each first conductive pattern is lower than the upper surface of the first insulating interlayer and faces the upper sidewall of the first metal pattern of the respective first contact plug in a horizontal direction.

8. The semiconductor device of claim 1, wherein each of the first conductive patterns has a sidewall slope such that a lower width of the first conductive pattern is less than an upper width of the first conductive pattern.

9. The semiconductor device of claim 1, wherein a distance in a horizontal direction between an uppermost sidewall of the first metal pattern of the respective first contact plug and the first conductive pattern is greater than a distance in the horizontal direction between the uppermost sidewall of the first barrier metal pattern of the respective first contact plug and the first conductive pattern.

10. The semiconductor device of claim 1, wherein the second insulating interlayer includes a silicon oxide-based material having a dielectric constant lower than a dielectric constant of the first insulating interlayer.

11. The semiconductor device of claim 1, further comprising a capping layer covering the plurality of first conductive patterns and further covering the second insulating interlayer.

12. A semiconductor device, comprising:

- a substrate;
- a first insulating interlayer on the substrate, the first insulating interlayer including a plurality of contact holes;
- a plurality of first contact plugs in the plurality of contact holes, respectively, each first contact plug including:
 - a first metal pattern; and
 - a first barrier metal pattern covering a portion of a sidewall of the first metal pattern and further covering a bottom of the first metal pattern, and an uppermost surface of the first barrier metal pattern being lower than an upper surface of the first metal pattern;
- a second insulating interlayer that is on the first insulating interlayer and is further on the plurality of first contact plugs; and
- a plurality of first conductive patterns passing through the second insulating interlayer and extending below an upper surface of the first insulating interlayer in a

vertical direction, and each of the plurality of first conductive patterns contacting an upper surface of a respective first contact plug of the plurality of first contact plugs and further contacting a respective upper portion of the first insulating interlayer,

wherein a distance in a horizontal direction between an uppermost sidewall of the first metal pattern of the respective first contact plug and the first conductive pattern is greater than a distance in the horizontal direction between an uppermost sidewall of the first barrier metal pattern of the respective first contact plug and the first conductive pattern.

13. The semiconductor device of claim **12**, wherein at least a portion of each first conductive pattern is lower than the upper surface of the first insulating interlayer and faces the first metal pattern of the respective first contact plug in a horizontal direction.

14. The semiconductor device of claim **12**, wherein the upper surface of the first insulating interlayer and the upper surface of the first metal pattern are coplanar with each other, and a first recess is between the first insulating interlayer and the first metal pattern.

15. The semiconductor device of claim **14**, wherein the second insulating interlayer occupies the first recess.

16. The semiconductor device of claim **12**, wherein the second insulating interlayer includes a silicon oxide-based material having a dielectric constant lower than a dielectric constant of the first insulating interlayer.

17. The semiconductor device of claim **12**, wherein each of the plurality of first conductive patterns includes a first bottom surface contacting a portion of an upper surface of the respective first contact plug, and a second bottom surface contacting the first insulating interlayer and being lower than the first bottom surface.

18. A semiconductor device, comprising:

- a substrate;
- a plurality of capacitors on a cell region of the substrate;
- a plurality of peripheral circuit patterns on a peripheral circuit region of the substrate;
- a first insulating interlayer covering the plurality of capacitors and further covering the plurality of peripheral circuit patterns;
- a plurality of first contact plugs passing through the first insulating interlayer and connecting to the plurality of peripheral circuit patterns, each of the plurality of first contact plugs including:

a first metal pattern; and

a first barrier metal pattern covering a portion of a sidewall of the first metal pattern and further covering a bottom of the first metal pattern, and the first barrier metal pattern further exposing an upper sidewall of the first metal pattern;

a second insulating interlayer that is on the first insulating interlayer and is further on the plurality of first contact plugs; and

a first conductive pattern passing through the second insulating interlayer and extending below an upper surface of the first insulating interlayer in a vertical direction, and the first conductive pattern contacting upper surfaces of the plurality of first contact plugs and further contacting an upper surface of the first insulating interlayer on the peripheral circuit region,

wherein a distance in a horizontal direction between an uppermost sidewall of the first metal pattern of a first contact plug of the plurality of first contact plugs and the first conductive pattern is greater than a distance in the horizontal direction between an uppermost sidewall of the first barrier metal pattern of the first contact plug and the first conductive pattern.

19. The semiconductor device of claim **18**, further comprising:

a plurality of second contact plugs passing through the first insulating interlayer and electrically connecting to respective capacitors of the plurality of capacitors, each of the plurality of second contact plugs including:

a first metal pattern; and

a first barrier metal pattern covering a portion of a sidewall of the first metal pattern and further covering a bottom of the first metal pattern, and the first barrier metal pattern further exposing an upper sidewall of the first metal pattern; and

second conductive patterns contacting an upper surface of each of the plurality of second contact plugs and contacting an upper surface of the first insulating interlayer on the cell region.

20. The semiconductor device of claim **18**, wherein the second insulating interlayer includes a silicon oxide-based material having a dielectric constant lower than a dielectric constant of the first insulating interlayer.

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